Computational Methods for a Global Solution of a Two-Country Model with Occasionally Binding Financial Constraints

A Theory-First Guide to the Algorithms, and to the Levers that Control Them

Technical companion to the QUANTITATIVE_MODEL repository

5 October 2026

What this document is. A self-contained account of the numerical mathematics behind the two solvers in this repository: a *global Chebyshev–Smolyak projection* solver for a nonlinear model with an occasionally binding incentive constraint, and a *sequence-space Jacobian* solver for its linearised sticky-price twin. The emphasis is roughly four-to-one on theory over implementation: every algorithm is stated as a mathematical object first, with its convergence properties, its failure modes, and the parameters ("levers") that move it. Code is referenced only as a pointer, in marked boxes, so that a reader can find the deployed counterpart of each construct without the document becoming a code walkthrough.

Who it is for. Early-stage doctoral students who know what a recursive competitive equilibrium is, have met value-function iteration and perturbation methods, and now need to understand what a production global solver actually does — and, more importantly, *why each design choice is the one that works*. Prior exposure to heterogeneous-agent macroeconomics and to financial-accelerator models is assumed; prior exposure to approximation theory is not.

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Part I The Computational Problem

1 What is being solved

1.1 The economic object

The model is a monetary union of two countries, D (small, risky) and F (large, safe). Each country has a continuum of households facing idiosyncratic income risk and a borrowing constraint; a representative firm with Cobb–Douglas technology; a capital producer with convex adjustment costs; a government issuing long-duration perpetuity debt under a fiscal feedback rule; and — the mechanism that makes the problem hard — a Gertler–Karadi/Bocola financial intermediary whose balance sheet is limited by an *incentive constraint* that binds only sometimes.

Write the intermediary's problem informally. A banker enters the period with net worth n, chooses a portfolio of assets A (capital, two sovereign bonds, working-capital loans), funds the gap $A - n$ with deposits, and may abscond with a fraction λ of the assets. Depositors lend only if the franchise value of continuing exceeds the diversion payoff:

$$\underbrace{\alpha(\mathbf{S})\, n}_{\text{franchise value}} \quad \geq \quad \underbrace{\lambda A}_{\text{diversion payoff}} \quad , \tag{1}$$

with multiplier $\mu \geq 0$ and complementary slackness $\mu\,[\alpha n - \lambda A] = 0$. When (1) is slack the banker is unconstrained and asset prices satisfy textbook Euler equations; when it binds, a wedge $\lambda\mu/\mathbb{E}[\Omega]$ opens between the return on intermediated assets and the deposit rate, and that wedge is the entire transmission channel from financial distress to real activity in this model.

1.2 Why this cannot be solved by perturbation alone

Three features of (1) defeat local methods.

1. **The constraint is occasionally binding.** The equilibrium conditions are not differentiable in the state: $\mu(\mathbf{S}) = \max\{\cdot, 0\}$ is C^0 but not C^1 at the switching boundary. A Taylor expansion around a point is an expansion around one *regime*; it cannot represent the switch. Worse, as Section 8.6 documents, this economy's ergodic rest point sits *on* the switching boundary, which is precisely where a smooth approximant is least reliable.

2. **Risk is priced, and pricing risk is a second-order object.** The mechanism of interest — an exogenous rise in the *priced* probability of sovereign default marks a long bond down, destroys bank equity, tightens (1), raises the lending spread and contracts output — lives in the covariance between the stochastic discount factor and the bond payoff. A first-order perturbation sets that covariance to its steady-state value by construction. One therefore needs either a higher-order expansion (which still cannot handle the kink) or a genuinely global method.

3. **The feared event is large.** The default branch is a 55% write-down on the whole claim. Nothing local survives a shock of that size: the post-default economy is not a small deviation from the pre-default one, and the *expectation* taken in today's bond price must integrate over both branches at their true distance apart.

The response is to solve the model *globally*: to approximate the equilibrium decision rules as explicit functions on a region of the state space, and to impose the equilibrium conditions at enough points in that region to pin the approximation down. That is the subject of Parts II–IV.

1.3 The functional equation

Collect the endogenous state in a vector $\mathbf{S} \in \mathcal{S} \subset \mathbb{R}^d$ and the period's unknown allocations in $\mathbf{x} \in \mathbb{R}^K$. A recursive competitive equilibrium is a collection of *decision rules* $g : \mathcal{S} \times \{0,1\} \to \mathbb{R}^K$ — one set per discrete regime, here the realised-default indicator $d' \in \{0,1\}$ — together with a law of motion $\mathbf{S}' = H(\mathbf{S}, \mathbf{x}, \epsilon')$, satisfying

$$\boxed{\mathcal{F}\big(\mathbf{S},\, d,\, g(\mathbf{S},d),\, g(\cdot,\cdot)\big) \;=\; \mathbf{0} \qquad \text{for all } \mathbf{S} \in \mathcal{S},\; d \in \{0,1\}.} \tag{2}$$

Here $\mathcal{F}$ is a vector of K equilibrium conditions — Euler equations, market clearing, first-order conditions, complementarity conditions — and the third argument appears *inside an expectation*: evaluating $\mathcal{F}$ at one state requires knowing g at every state reachable next period. Equation (2) is therefore a *functional* equation: the unknown is a function, and the equation must hold at a continuum of points.

Two approximations convert (2) into something a computer can solve:

Approximate the function. Replace g by $\hat{g}(\cdot\,;\boldsymbol{\theta})$, a member of a finite-dimensional family indexed by coefficients $\boldsymbol{\theta}$. Part II is about choosing that family well.

Approximate "for all S". Impose (2) only at a finite set of *collocation points* $\{\mathbf{S}_i\}_{i=1}^n \subset \mathcal{S}$, chosen so that the finite system has as many equations as unknowns. Part III is about solving the resulting finite system.

1.4 Two solvers, one model, different jobs

The repository carries two independent solvers and they are not substitutes.

	Global projection	**Sequence-space (SSJ)**
Mathematical object	recursive decision rules on a region of the state space	impulse-response Jacobian around a point
Nonlinearity	fully nonlinear; the constraint switches	linearised; the constraint's regime is fixed
Unknown	$\boldsymbol{\theta} \in \mathbb{R}^{4600}$ (policy values at nodes)	$\mathbf{U} \in \mathbb{R}^{27T}$ (unknown paths)
Solved by	one damped Newton on the whole coefficient vector	one linear solve $G = -H_U^{-1} H_Z$
Cost	hours	minutes
Carries	occasionally binding IC, priced default, two regimes	sticky prices, nominal deposit contracts, the paper's figures

The two are calibrated differently and their impulse magnitudes are not comparable. Parts II–VI concern the global solver; Part VII treats the sequence-space method, whose mathematics is entirely different and worth knowing in its own right.

In the code. `run.py` is the single entry point and holds the solver settings (grid resolution, which experiments run); the economics live in `calibration/`. The global pipeline is `compute_global()`/`plot_global()` in `run.py`; the linear pipeline is `linear_ssj/main.py`. The two need different Python interpreters.

Part II Approximating Functions of Many Variables

2 Chebyshev interpolation in one dimension

Everything in Part II is built from the one-dimensional case, so it pays to understand that case properly.

2.1 The question approximation theory actually answers

The Weierstrass theorem says that any continuous f on $[-1,1]$ is uniformly approximable by polynomials. It is useless operationally: it does not say which polynomial, and it does not say how the error falls with degree. The constructive questions are: *given m evaluations of f, where should they be taken, and what is the resulting error?*

Fix the family: polynomials of degree $\leq m-1$, written in the *Chebyshev basis*

$$T_k(u) = \cos\big(k \arccos u\big), \qquad u \in [-1,1], \qquad k = 0,1,2,\ldots \tag{3}$$

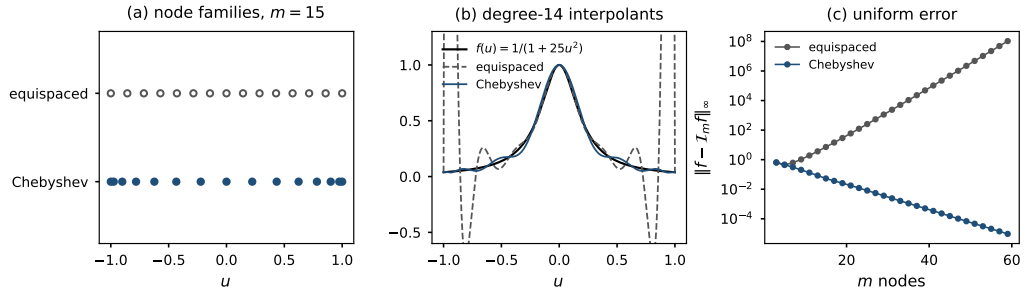


Figure 1: Node placement is not a detail. Panel (c) is computed, not schematic: equispaced interpolation of a smooth function diverges, Chebyshev interpolation of the same function converges geometrically. The ratio of the two Lebesgue constants at $m = 40$ exceeds 10^9.

These satisfy the three-term recurrence

$$T_0(u) = 1, \quad T_1(u) = u, \quad T_k(u) = 2u\,T_{k-1}(u) - T_{k-2}(u), \tag{4}$$

which is how they are evaluated in practice: (4) is $O(m)$ per point, backward-stable, and needs no special functions.

> **In the code.** `chebyshev_basis_1d(x, max_deg)` in `global_projection/solver_recursive/state_grid.py` is exactly (4), filling an (points) × (max_deg + 1) matrix column by column.

2.2 Why node placement is the whole game

Let $\mathcal{I}_m f$ denote the degree-$(m-1)$ polynomial interpolating f at nodes $u_1 < \cdots < u_m$. The error splits into a *best-approximation* part and an *amplification* part:

$$\|f - \mathcal{I}_m f\|_\infty \;\le\; \big(1 + \Lambda_m\big) \inf_{p \in \mathcal{P}_{m-1}} \|f - p\|_\infty, \tag{5}$$

where $\Lambda_m = \big\| \sum_{j=1}^{m} |\ell_j| \big\|_\infty$ is the *Lebesgue constant* of the node set and ℓ_j are the Lagrange basis polynomials. Equation (5) is the key organising fact: the best polynomial may be excellent, but interpolation at badly chosen nodes amplifies its error by Λ_m, and

$$\text{equispaced nodes:} \quad \Lambda_m \sim \frac{2^m}{e\, m \log m} \quad \text{(exponential)}, \tag{6}$$

$$\text{Chebyshev nodes:} \quad \Lambda_m \sim \frac{2}{\pi} \log m + O(1) \quad \text{(logarithmic)}. \tag{7}$$

An exponentially growing amplification factor is what produces the *Runge phenomenon*: wild oscillation near the endpoints, worsening with degree, even for a perfectly smooth f. Figure 1 shows all three facts — the node families, the resulting degree-14 interpolants of Runge's function $f(u) = 1/(1 + 25u^2)$, and the uniform error against m. The equispaced error *diverges*; the Chebyshev error falls geometrically to machine precision.

2.3 Which Chebyshev nodes, and why these ones

Two node families carry the name. The *Chebyshev points of the first kind* (zeros of T_m) are $\cos\big(\pi(2j-1)/2m\big)$; the *points of the second kind*, also called Chebyshev–Lobatto or Chebyshev *extrema*, are

$$u_j^{(m)} = -\cos\Big(\frac{\pi(j-1)}{m-1}\Big), \qquad j = 1, \ldots, m. \tag{8}$$

Both have logarithmic Lebesgue constants. The solver uses (8), for a reason that becomes decisive in Section 3: the extrema are *nested*. Taking $m_i = 2^{\,i-1} + 1$ points at "level" i gives

$$\mathcal{U}_1 = \{0\} \subset \mathcal{U}_2 = \{-1, 0, 1\} \subset \mathcal{U}_3 = \{-1, -\tfrac{\sqrt{2}}{2}, 0, \tfrac{\sqrt{2}}{2}, 1\} \subset \cdots \tag{9}$$

so that every node of a coarse level survives into every finer level. Nesting is what makes the sparse-grid construction economical, and it is also what makes *refinement a warm start*: a solution computed on level i is exactly reusable at level $i{+}1$.

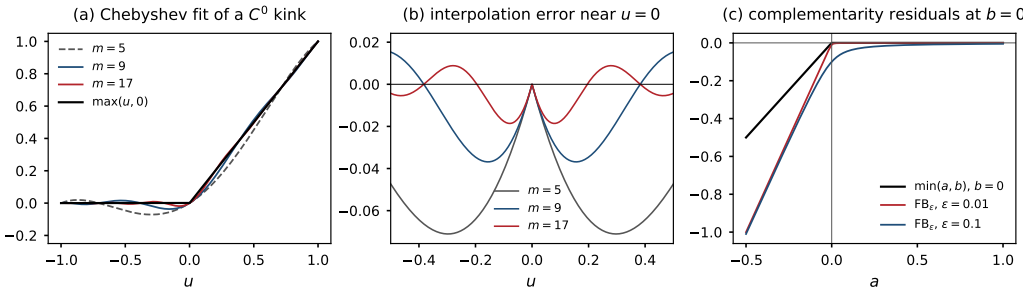


Figure 2: Nonsmoothness, computed. (a) Chebyshev interpolants of $\max(u, 0)$ at three resolutions: the fit returns strictly positive values where the truth is exactly zero, and negative values just left of the kink. (b) The error narrows but does not shrink in amplitude — a Gibbs phenomenon. (c) Two ways of writing complementarity: $\min(a, b)$ has *zero slope* in a whenever $b < a$, whereas the smoothed Fischer–Burmeister function retains slope in both arguments (Section 8).

> **In the code.** `_level_points(i)` builds $\mathcal{U}_i$; `_new_points(i)` returns $\mathcal{U}_i \setminus \mathcal{U}_{i-1}$, which for $i \geq 3$ is just the odd-indexed entries (`[1::2]`) — a one-line consequence of nesting.

2.4 Error bounds: the smooth case and the kinked case

Two regimes of convergence matter here, and the model lives in both.

Theorem 2.1 (Geometric convergence for analytic functions)**.** *If f extends analytically to the Bernstein ellipse E_ρ with foci ± 1 and semi-axis sum $\rho > 1$, and $|f| \leq M$ on E_ρ, then*

$$\|f - \mathcal{I}_m f\|_\infty \;\leq\; \frac{4M}{\rho - 1}\,\rho^{-(m-1)}. \tag{10}$$

The error falls *geometrically* in the number of nodes, at a rate set by how far f's nearest singularity sits from $[-1, 1]$. This is the regime the solver wants to be in, and it explains the shape of Figure 1(c).

Theorem 2.2 (Algebraic convergence for finitely differentiable functions)**.** *If $f \in C^k[-1, 1]$ then $\|f - \mathcal{I}_m f\|_\infty = O(m^{-k} \log m)$. For $f \in C^0$ with a single interior kink — the case $\mu = \max\{\cdot, 0\}$ — the rate is $O(m^{-1})$ and the error does not decay* uniformly*: it concentrates in a shrinking neighbourhood of the kink, where it is $O(1)$ in a relative sense. This is the* Gibbs phenomenon*.*

Figure 2 panels (a) and (b) show this directly for $\max(u, 0)$. Note the qualitative behaviour: the overshoot does not vanish as m grows, it merely narrows and migrates toward the kink. *Refinement does not cure a Gibbs phenomenon; it relocates it.* Section 8.6 shows what this costs the model, and what the honest response is.

2.5 Collocation versus regression, and why squareness matters

Given n basis functions and n nodes, interpolation solves the square linear system

$$\Phi\, \boldsymbol{c} = \boldsymbol{y}, \qquad \Phi_{ij} = \phi_j(u_i), \tag{11}$$

and the resulting $\hat{f} = \sum_j c_j \phi_j$ reproduces f *exactly at the nodes*. With more nodes than basis functions one would instead run a least-squares regression, which does not. The distinction is not cosmetic in this application. The global solver treats *the policy values at the nodes* as its unknowns and reads the continuation value off the fitted interpolant; if the interpolant did not reproduce its own nodes, the "value at a node" would not be well-defined and the Newton system would be inconsistent with its own parameterisation. Squareness is thus a structural requirement, not an efficiency choice.

One practical consequence: Φ is fixed once the grid is built, so it is factorised once (an LU) and every subsequent fit is a triangular solve.

> **In the code.** `SmolyakGrid.__init__` builds Φ and stores `lu_factor(_Phi)`; `fit()` is one `lu_solve`. A separate `fit_weighted()` offers masked ridge-regularised least squares — deliberately *not* the default, because it breaks nodal exactness (see the comment in `_install_guess`).

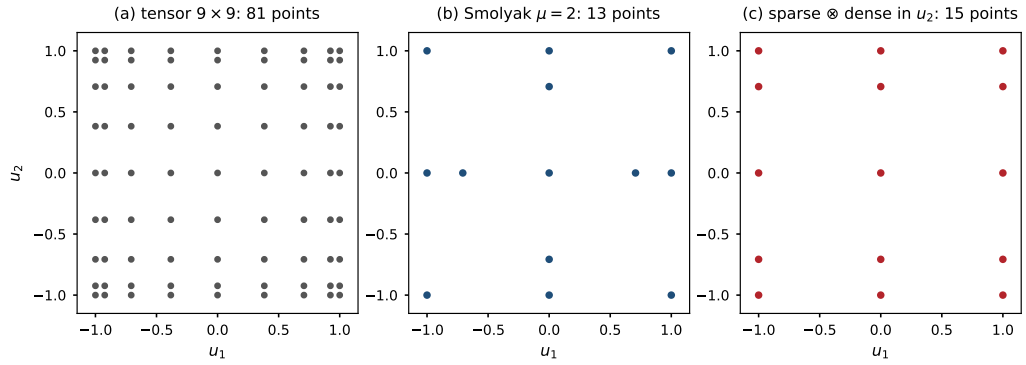


Figure 3: Grid geometry in two dimensions, drawn by the repository's own grid constructor. (a) The tensor product is dense everywhere. (b) The Smolyak grid at $\mu = 2$ keeps the axes and the coarse interactions, dropping the high-order corners. (c) The construction actually shipped: a sparse factor over all dimensions but one, tensored with a *dense* factor in the dimension that carries curvature — here u_2 stands for the sovereign-risk state s.

3 The curse of dimensionality and the Smolyak construction

3.1 Why tensor products are hopeless at $d = 12$

The obvious extension of §2 to d dimensions is the tensor product: take m nodes per dimension, use all m^d combinations, and all m^d product basis functions $\phi_{\boldsymbol{k}}(\boldsymbol{u}) = \prod_j T_{k_j}(u_j)$. The approximation inherits one-dimensional accuracy in every direction. It also inherits an exponential cost: at the model's $d = 12$ with a modest $m = 5$,

$$m^d = 5^{12} \approx 2.44 \times 10^8 \text{ points.} \tag{12}$$

Each point requires an evaluation of the full period map including multi-branch quadrature. This is not a tractability-at-the-margin problem; it is off by eight orders of magnitude.

3.2 The telescoping-difference idea

Smolyak's construction rests on one observation: most of the tensor-product basis is paying for *high-order interaction terms* that a smooth function does not need. A function that is quadratic in each variable separately and has mild pairwise interaction does not require the degree-$(4, 4, 4, \ldots, 4)$ cross terms that the tensor grid buys.

Formalise it. Let $\mathcal{I}^{(i)}$ be the one-dimensional interpolation operator at level i (using $\mathcal{U}_i$), and define the *difference operator*

$$\Delta^{(i)} \;=\; \mathcal{I}^{(i)} - \mathcal{I}^{(i-1)}, \qquad \mathcal{I}^{(0)} \equiv 0. \tag{13}$$

Then $\mathcal{I}^{(i)} = \sum_{\iota \le i} \Delta^{(\iota)}$ telescopes, and the full d-dimensional tensor operator at level i in every dimension is

$$\mathcal{I}^{(i)} \otimes \cdots \otimes \mathcal{I}^{(i)} = \sum_{\boldsymbol{\iota}\,:\,\iota_j \le i \;\forall j} \Delta^{(\iota_1)} \otimes \cdots \otimes \Delta^{(\iota_d)}. \tag{14}$$

The sum runs over a d-dimensional *cube* of multi-indices. Smolyak's operator keeps only a *simplex* of them:

$$\boxed{\mathcal{A}_{\mu,d} \;=\; \sum_{\boldsymbol{\iota}\,:\,\sum_j (\iota_j - 1) \le \mu} \Delta^{(\iota_1)} \otimes \cdots \otimes \Delta^{(\iota_d)}.} \tag{15}$$

The parameter μ is a *total-level budget*: a term is admitted if the levels it uses, summed across dimensions, do not exceed μ above the baseline. High resolution in one dimension is affordable; high resolution in all dimensions simultaneously is not. The discarded terms are exactly the high-order interactions.

Proposition 3.1 (Point count)**.** *With nested node sets, the Smolyak grid is* $\mathcal{H}_{\mu,d} = \bigcup_{\boldsymbol{\iota}} \left(\mathcal{U}_{\iota_1}^{\text{new}} \times \cdots \times \mathcal{U}_{\iota_d}^{\text{new}}\right)$ *where* $\mathcal{U}_i^{\text{new}} = \mathcal{U}_i \setminus \mathcal{U}_{i-1}$*. For* $\mu = 1$ *the count is* $1 + 2d$ *— the centre plus two endpoints per dimension. For* $\mu = 2$ *it grows as* $O(d^2)$*, and in general* $|\mathcal{H}_{\mu,d}| = O(d^\mu)$*:* polynomial in dimension at fixed budget*, against the tensor grid's exponential.*

At $d = 12$: $\mu = 1$ gives 25 points, $\mu = 2$ gives 313. The problem has gone from 2.4×10^8 to a few hundred. This is the single most important fact in the document.

3.3 The square "new points, new degrees" form

Written as (15), the Smolyak operator is a signed combination of tensor interpolants, and implementing it that way means carrying combinatorial coefficients and re-evaluating overlapping terms. The implementation here uses the equivalent *Krueger–Kübler* form, which is cleaner and has a property worth naming.

The idea: pair each *new point set* $\mathcal{U}_i^{\text{new}}$ with a matching set of *new Chebyshev degrees* $\mathcal{D}_i^{\text{new}}$, constructed so that $|\mathcal{D}_i^{\text{new}}| = |\mathcal{U}_i^{\text{new}}|$:

$$\mathcal{D}_1^{\text{new}} = \{0\}, \quad \mathcal{D}_2^{\text{new}} = \{1, 2\}, \quad \mathcal{D}_i^{\text{new}} = \{2^{i-2} + 1, \ldots, 2^{i-1}\} \ (i \geq 3). \tag{16}$$

Running the *same* multi-index set $\{\boldsymbol{\iota} : \sum_j (\iota_j - 1) \leq \mu\}$ over degrees instead of points yields exactly as many basis functions as nodes. The interpolation matrix Φ is therefore square, the collocation system (11) is a plain linear solve, and the interpolant is exact at every node — the property §2 argued is structural.

Proposition 3.2 (Polynomial exactness)**.** *$\mathcal{A}_{\mu,d}$ reproduces exactly every polynomial in the span of $\{\prod_j T_{k_j} : \boldsymbol{k} \in \mathcal{D}_{\mu,d}\}$. At $\mu = 1$ that span contains all constants, all linear terms, and the pure quadratics u_j^2 — but* no *cross terms $u_j u_k$. At $\mu = 2$ it contains all quadratic cross terms.*

This proposition is the practical guide to choosing μ: ask which interactions the economics requires. If the only strongly curved object is a function of one state, $\mu = 1$ plus refinement in that state suffices. If two states interact nonlinearly, $\mu \geq 2$ is not optional.

In the code. `_multi_indices(d, mu, mu_vec)` enumerates the simplex depth-first; `_new_degrees(i)` is the degree set above; `_sparse_nodes` takes the union of tensor products of new-point sets and records the matching degrees. The exactness claim is regression-tested in `Claude files/global_projection/tests/test_state_grid.py`. An independent check: fitting $1 + u_1 u_9 + u_9^2$ on the shipped grid and evaluating at 200 random interior points reproduces it to 8.9×10^{-16}.

3.4 Anisotropy: per-dimension budgets

Nothing forces the budget to be isotropic. Replacing the constraint set by

$$\left\{\boldsymbol{\iota} : \sum_j (\iota_j - 1) \leq \mu \ \text{ and } \ \iota_j - 1 \leq \mu_j \ \forall j\right\} \tag{17}$$

caps the resolution dimension by dimension. This is the right instrument when the state vector is heterogeneous in curvature, which it invariably is: in this model the two capital stocks are nearly linear in their own arguments, while the risk state s drives a logistic default probability.

3.5 The refinement that is actually shipped: sparse ⊗ dense

Anisotropic budgets have a defect. Because the constraint $\sum_j (\iota_j - 1) \leq \mu$ is *global*, raising one dimension's cap to reach degree 4 in that dimension requires raising μ, which then admits high-level terms in *every other* dimension too. On the 10-state box, asking for level 2 in the risk dimension alone costs 165 points; the isotropic $\mu = 2$ grid costs 221.

The alternative is to make the grid a *Cartesian product of two interpolation operators*:

$$\boxed{\mathcal{A} \;=\; \mathcal{A}_{\mu,\,d-1}^{(\text{all dims but } r)} \;\otimes\; \mathcal{I}_{(r)}^{m}} \tag{18}$$

where $\mathcal{I}_{(r)}^{m}$ is a *dense* m-node Chebyshev interpolant in the single dimension r. Both factors are square interpolation operators, so the product is square and exact at its nodes — the structural property survives. What one buys is degree $m - 1$ in dimension r *and full interaction* between that dimension and the entire sparse basis, at a cost of $|\mathcal{H}_{\mu,d-1}| \times m$ points.

The measured payoff on this model's curvature profile (Figure 4) is large. Against the isotropic $\mu = 2$ grid at 221 points and relative RMS error 3.9×10^{-2}, the product grid at $m = 9$ achieves 1.1×10^{-3} — about 36 times more accurate — using 171 points, i.e. *fewer*. The reason is specific and instructive: the curvature being resolved is the logistic $p^d(s) = (1 + e^{-s})^{-1}$, which a $\mu = 1$ quadratic misfits badly (reading 3.30% where the truth is 1.98% at the headline shock), and dimension $r = s$ is exactly where the extra degrees go.

In the code. `SmolyakGrid(..., refine=(dim, m))` implements (18) via `_tensor_refine`. The shipped setting is `S_REFINE = 5` in `run.py` (115 points on the 12-state box); `S_REFINE = 9` is the reference resolution (207 points) and the solve ladder walks $5 \to 9$. Confirmed convergence at 5: moving to 9 shifts the impact output response by 1.6%–3.4% of itself, well inside the reported identification bracket, and every other headline number moves only in the third digit.

4 Domain geometry: where the approximation lives

A global method is global only on a *region*. Choosing that region is a modelling decision with real consequences, and it is governed by three considerations: coverage, invariance, and coordinates.

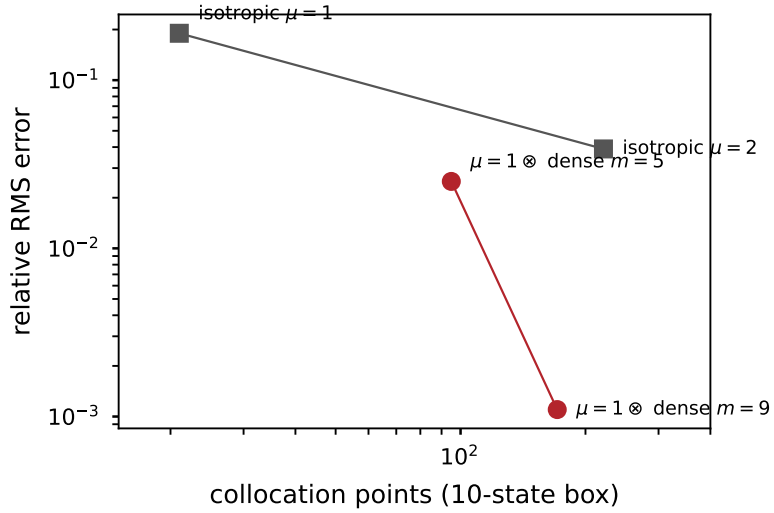


Figure 4: Measured relative RMS error against point count, on a test function matching this model's curvature profile (near-linear in the wealth states, logistic in s, with one interaction). The product construction (18) dominates: it is more accurate than the isotropic $\mu = 2$ grid at fewer points, because its extra degrees are spent where the curvature is.

4.1 The affine map and the coverage question

The grid is constructed on $[-1, 1]^d$ and mapped affinely onto a box $\prod_j [\underline{S}_j, \overline{S}_j]$:

$$\boldsymbol{u} = 2\,\frac{\mathbf{S} - \underline{\mathbf{S}}}{\overline{\mathbf{S}} - \underline{\mathbf{S}}} - \mathbf{1}, \qquad \mathbf{S} = \underline{\mathbf{S}} + \tfrac{1}{2}(\boldsymbol{u} + \mathbf{1}) \odot (\overline{\mathbf{S}} - \underline{\mathbf{S}}). \tag{19}$$

The box must be wide enough that the model's ergodic set lies inside it (otherwise the solver is extrapolating, and polynomial extrapolation is catastrophic), and narrow enough that the approximation is not spending its degrees of freedom on states the model never visits.

For a stochastic state the right unit is *coverage in unconditional standard deviations*, not an absolute width. If s_t is an AR(1) with persistence ρ_s and innovation standard deviation σ_s, its unconditional standard deviation is $\sigma_s/\sqrt{1-\rho_s^2}$, and the box is set to $s^\star \pm \kappa\,\sigma_s/\sqrt{1-\rho_s^2}$ with κ the coverage. Stating the lever this way makes it invariant to recalibration of σ_s; an absolute half-width silently changes coverage whenever σ_s moves.

Remark 4.1 (A cautionary measurement). There is a documented episode in which an absolute half-width, chosen when σ_s was 2.4 times its eventual value, left the box only 1.4–2.3 *conditional* standard deviations wide. The consequence was not a mild loss of accuracy: because simulated states are projected back into the box, the clipping mean-reverted s harder than ρ_s did, cutting the *effective* persistence from 0.95 to 0.80 and removing roughly 62% of the long bond's mark-to-market repricing — the paper's entire mechanism. A box is part of the model, not part of the plumbing.

4.2 One-step invariance and the non-normality obstruction

Ideally the box is *forward invariant*: states reachable in one step from inside the box remain inside. Linearise the law of motion for a sub-block of states around the steady state, $\mathbf{S}' - \mathbf{S}^{ss} \approx J(\mathbf{S} - \mathbf{S}^{ss})$. A box with half-widths $\boldsymbol{b}$ is one-step invariant if

$$|J|\,\boldsymbol{b} \;\le\; \boldsymbol{b} \qquad \text{componentwise,} \tag{20}$$

where $|J|$ is the entrywise absolute value. By Perron–Frobenius, (20) has a positive solution $\boldsymbol{b}$ if and only if the spectral radius $\rho(|J|) \le 1$. Crucially $\rho(|J|)$ is *not* $\rho(J)$: for a non-normal J they can differ a great deal.

This model supplies a textbook example. The measured Jacobian of the two banks' gross deposit obligations (P_D, P_F) at the steady state is

$$J = \begin{pmatrix} 1.076 & -0.310 \\ 2.016 & -1.252 \end{pmatrix}, \qquad \text{eigenvalues } 0.766,\ -0.942, \qquad \rho(|J|) = 1.96. \tag{21}$$

The dynamics are *stable* ($\rho(J) < 1$) and yet *no axis-aligned box centred on the steady state is one-step invariant, at any bandwidth*, because $\rho(|J|) > 1$. The cause is non-normality: the singular values are 2.61 and 0.28, so the map stretches strongly in one direction while contracting in another, and an axis-aligned box cannot align with either.

4.3 Rotation: collocating on the eigenbasis

The repair is to draw the box in coordinates where J is diagonal. Let $J = V\Lambda V^{-1}$ and collocate on $\boldsymbol{z} = V^{-1}(\mathbf{S} - \mathbf{S}^c)$. In those coordinates the map acts component-wise with $|\lambda_j| < 1$, so *every* box is invariant. The grid class supports this by carrying a general invertible rot and centre, converting on the way in and out, so that every public method still speaks natural coordinates.

Converting a box specification between coordinate systems requires care. A $\boldsymbol{z}$-box with half-widths $\boldsymbol{b}$ reaches natural half-widths $|V|\,\boldsymbol{b}$, so to reach prescribed natural half-widths $\boldsymbol{t}$ one solves $|V|\boldsymbol{b} = \boldsymbol{t}$ — and falls back to the superset $\boldsymbol{b} = |V|\boldsymbol{t}$ if that has no positive solution.

Remark 4.2 (Theory wins, measurement decides). The rotation is *right in theory* and *measures worse*, so it ships switched off. A rotated box is a parallelogram in natural coordinates, and its corners reach combinations of the two deposit obligations at which the period map has no solution. The invariance gained is worth less than the feasibility lost. This is a recurring pattern in global solution work and worth internalising early: the correct geometric argument tells you what to try, not what to keep.

4.4 Shear: coordinates on the reachable set

A second, subtler coordinate problem arises when two states are linked by an *accounting identity* rather than by dynamics. The central bank's backstop in this model adds two states: its bond holdings M and its gross obligation O to the banks. A purchase moves bank holdings b_{DD} and M one-for-one and books $O \approx R\,Q\,M$. The three are therefore tied, and an axis-aligned box in (b_{DD}, M, O) contains nodes that *no purchase can reach*: bonds sold with nothing owed, a claim owed with nothing held.

Collocating at unreachable nodes is not a harmless waste. The fitted rule must interpolate through those nodes, and the measured consequence was a *non-monotone* price response to a forced purchase (-0.5% at one size, $+0.3\%$ at another) and a fourfold move in the constraint multiplier. The repair is a *shear* — a unit-determinant change of coordinates putting the box on the reachable directions:

$$z_b = b_{DD} + M, \qquad z_O = O - \rho\,M, \qquad \rho \equiv (1 + r^{dep})\,Q^{ss}, \tag{22}$$

i.e. "what the bank would hold absent the backstop" and "the claim net of its steady-state value". Because both new states are zero at the steady state and the shear is unit-triangular, the $M = O = 0$ slice is untouched: every node of the smaller grid remains a node of the larger one, so the backstop-off model *nests* the no-backstop solution exactly rather than approximately (measured agreement of the rules: 5.5×10^{-10}).

> **In the code.** `build_state_box` composes the optional P-rotation with the backstop shear (they touch disjoint rows) and hands the product to `SmolyakGrid(rot=...)`. `_rotate_box` performs the half-width conversion; `outside()` reports per-state box violation as a fraction of width, which is the invariance diagnostic of §14. Nesting is asserted in `Claude files/global_projection/tests/test_recursive_nesting.py`.

Part III Solving the Discretised System

5 Two paradigms: iterate, or solve

With the function space fixed, (2) becomes a finite system. There are two fundamentally different ways to attack it, and the difference matters far more than it looks.

5.1 Time iteration as a fixed-point problem

The traditional approach constructs an operator. Fix a guess $\hat{g}^{(k)}$ for the decision rules. At each collocation point $\mathbf{S}_i$, treat $\hat{g}^{(k)}$ as the *frozen continuation* — the function used to evaluate next period inside the expectation — and solve the K period-t equilibrium conditions for the K unknowns at that point:

$$\mathbf{x}_i^{\text{new}}: \quad \mathcal{F}\big(\mathbf{S}_i,\, d,\, \mathbf{x}_i^{\text{new}},\, \hat{g}^{(k)}\big) = \mathbf{0}. \tag{23}$$

Then fit a new interpolant through $\{\mathbf{x}_i^{\text{new}}\}$, optionally damp, and repeat:

$$\hat{g}^{(k+1)} = (1-\omega)\,\hat{g}^{(k)} + \omega\,\mathcal{T}\big[\hat{g}^{(k)}\big]. \tag{24}$$

This is attractive. Each pointwise solve is small ($K = 14$ unknowns here, not 4600), the points are independent, and if $\mathcal{T}$ is a contraction the iteration converges globally from a crude start. It is the workhorse of the literature.

5.2 The contraction rate is the problem

Suppose $\mathcal{T}$ is a contraction with modulus ρ. Convergence is *linear*: $\|\hat{g}^{(k)} - g\| \lesssim \rho^k$. Two consequences follow, and both bite here.

Proposition 5.1 (Level error implied by a change-based stopping rule)**.** *If the iteration converges at rate ρ and one stops when the* change $\|\hat{g}^{(k+1)} - \hat{g}^{(k)}\|$ *falls below τ, the remaining* level *error is*

$$\|\hat{g}^{(k)} - g\| \;\approx\; \frac{\tau}{1-\rho}. \tag{25}$$

At $\rho = 0.99$, a stopping tolerance of $\tau = 10^{-6}$ admits a level error of 10^{-4}. That is not a tight solution, and nothing in the diagnostics reveals it.

Now the measured ρ for this model. The binding mode is the banker's franchise-value recursion

$$\alpha = \frac{\mathbb{E}[\Omega]\, R}{1-\mu}, \qquad \Omega = \beta\big[f + (1-f)\,\alpha'\big], \tag{26}$$

whose slope in α' is $\beta(1-f)R/(1-\mu) \approx 0.96$. With damping $\omega = 0.25$ the effective per-sweep modulus is $1 - \omega(1 - 0.96) \approx 0.990$, i.e. roughly 235 sweeps per decade of rule change. The symptom observed in practice is diagnostic and should be recognised: runs reported $\max|\mathcal{F}| = 10^{-14}$ *and* "rule-change tolerance not reached" simultaneously. Every collocation point was clearing *perfectly against a continuation that was still moving.* The residual, the natural object to monitor, was silent about the error.

5.3 Simultaneous collocation: make the policy values the unknowns

The alternative abandons the operator view. Define the unknown vector to be the policy *values at every node, in every regime*:

$$\boldsymbol{\theta} \;=\; \Big(\hat{x}_k(\mathbf{S}_i, d)\Big)_{k=1,\ldots,R;\; d=0,1;\; i=1,\ldots,n} \;\in\; \mathbb{R}^{R\cdot 2\cdot n}, \tag{27}$$

and define a single global residual map $F : \mathbb{R}^{R\cdot 2\cdot n} \to \mathbb{R}^{N_{\text{res}}\cdot 2\cdot n}$ by:

> *(i)* fit the Chebyshev coefficients to $\boldsymbol{\theta}$; *(ii)* at every $(\mathbf{S}_i, d)$, walk the period map using *that* interpolant as the continuation; *(iii)* stack the equilibrium residuals.

Then solve $F(\boldsymbol{\theta}) = \mathbf{0}$ by Newton's method. There is no inner root-find and no frozen continuation: *the continuation is always the current guess.* The fixed point is not approached at some rate — the residual itself is driven to its arithmetic floor, and there is no contraction modulus for error to leak through.

The two paradigms are compared below.

	Time iteration	**Simultaneous collocation**
Unknowns per solve	$K = 14$, repeated $2n$ times	$R \cdot 2n = 4600$, once
Continuation	previous iterate (frozen)	the current guess (self-consistent)
Convergence	linear at $\rho \approx 0.99$	quadratic in a neighbourhood
Stopping on	rule change (admits $\tau/(1-\rho)$)	the residual itself
Cost per step	$2n$ small root-finds	one dense Jacobian: 4601 residual evaluations
Role here	**warm start only**	**the solver**

5.4 The subtle part: self-reference and the identity residuals

Making *everything* an unknown requires care about objects that the pointwise method used to *read off* a recursion rather than solve for. In this model six such objects exist: the two banker franchise values α_D, α_F, the two consumptions C_D, C_F, and the two working-capital rates r_D^{wc}, r_F^{wc}. In time iteration these are computed from the frozen continuation, stored, and refitted — which is exactly the knot Proposition 5.1 describes.

Under simultaneous collocation they become unknowns with their own residual. The device is to impose consistency between the *guessed* value and the *implied* value:

$$\boxed{\; r_k \;=\; \mathcal{T}_k\big(\text{guess}_k\big) - \mathcal{T}_k\big(\text{implied}_k\big) \;=\; \log\frac{\text{guess}_k}{\text{implied}_k} \;} \tag{28}$$

where $\mathcal{T}_k$ is whatever transform the rule is fitted in (log level, or log gross rate; see §12). The logarithm is not cosmetic: it makes the residual scale-free and dimensionally comparable across objects of very different magnitudes.

The reason this closes the knot exactly is worth stating, because it is the link back to §2: *square collocation interpolates its own nodes.* When the period map asks for α at the current state and that state *is* a collocation

node, evaluating the fitted interpolant returns the guess exactly. There is no interpolation error at the node to contaminate (28), and the solved object is a genuine recursive equilibrium rather than a fixed point of a damped map.

Counting for this model: 14 market-clearing/Euler/policy residuals from the period map, plus 6 identity residuals, for $N_{\text{res}} = 20$ per (point, regime) — matching $R = 20$ stored rules, so the system is square.

> **In the code.** `collocation.py` is the solver. `make_residual` builds $F(\boldsymbol{\theta})$ and is the image of Bocola's `residual_model.m`; `pack`/`unpack` define the stacking convention (rule-major, then regime, then point); `point_residuals` in `point_map.py` is the per-point period map and is *unchanged* between the two paradigms — only who calls it and what counts as an unknown differs. The log in (28) is `to_fit(k, guess) - to_fit(k, out[k])`.

6 The Newton solver

6.1 Damped Newton with a finite-difference Jacobian

The iteration is

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - c\left[J(\boldsymbol{\theta}^{(k)})\right]^{-1} F(\boldsymbol{\theta}^{(k)}), \qquad J_{ij} = \frac{\partial F_i}{\partial \theta_j}, \tag{29}$$

with damping $c \in (0, 1]$. The Jacobian is built by forward differences, column by column:

$$J_{:,j} \;=\; \frac{F(\boldsymbol{\theta} + \varepsilon\, \boldsymbol{e}_j) - F(\boldsymbol{\theta})}{\varepsilon}. \tag{30}$$

No analytic Jacobian is available — F contains root-finds, interpolations, quadrature and smoothed max operators — and an automatic-differentiation pass through all of that would be a substantial engineering project for a derivative that (30) supplies adequately.

The step ε is a genuine lever with a two-sided error. Truncation error in (30) is $O(\varepsilon)$; floating-point cancellation error is $O(\epsilon_{\text{mach}}\|F\|/\varepsilon)$. Balancing gives $\varepsilon^\star \sim \sqrt{\epsilon_{\text{mach}}} \approx 1.5 \times 10^{-8}$ for an $O(1)$ function, but the relevant scale here is set by the *smoothness* of F, not by machine precision: F contains smoothed kinks with smoothing width η, and (30) is only a valid derivative if $\varepsilon \ll \eta$ is false — i.e. one needs $\eta \gtrsim \varepsilon$, so the smoothing must be *wider* than the differencing step. This interaction is the subject of §8.3; the shipped values are $\varepsilon = 10^{-6}$ and $\eta = 10^{-4}$.

6.2 Convergence theory, and what it does and does not promise

Theorem 6.1 (Newton–Kantorovich, informally)**.** *Let F be continuously differentiable near a root $\boldsymbol{\theta}^\star$ with $J(\boldsymbol{\theta}^\star)$ nonsingular and J Lipschitz with constant L. Then there is a neighbourhood of $\boldsymbol{\theta}^\star$ on which the undamped Newton iteration converges, and*

$$\|\boldsymbol{\theta}^{(k+1)} - \boldsymbol{\theta}^\star\| \;\le\; \frac{L\,\|J(\boldsymbol{\theta}^\star)^{-1}\|}{2}\,\|\boldsymbol{\theta}^{(k)} - \boldsymbol{\theta}^\star\|^2 : \tag{31}$$

the error squares *each step.*

Two features of this theorem organise everything in the rest of Part III.

It is local. The guarantee holds in a neighbourhood whose size depends on L and on $\|J^{-1}\|$, neither of which is known in advance. Outside that basin Newton can diverge, oscillate, or step into a region where F cannot be evaluated at all. Everything in §7 exists to deliver a starting point inside the basin.

It says nothing about uniqueness. Newton finds *a* root. In a model with an occasionally binding constraint there are genuinely nearby allocations — the binding and the slack equilibrium — and a Newton step can slip from one to the other. This is not a hypothetical: re-solving the period map at each simulated state, warm-started at the fitted rules, was measured to slip onto the slack branch in 49 of 60 periods, with the multiplier walking from 0.103 to 0 and the entire impulse response flipping sign (output +2.5% on impact against −0.004% for the rule-consistent read). The repair is to read the rules rather than re-solve, which is why §13 insists on it.

6.3 Globalisation: line search, stalling, and the arithmetic floor

Pure Newton is globalised by a backtracking line search on the merit function $\varphi(\boldsymbol{\theta}) = \frac{1}{2}\|F(\boldsymbol{\theta})\|_2^2$: try the full damped step, and halve the damping until φ decreases. This is standard. What is *not* standard, and is specific to models of this kind, is the stopping rule.

The original routine stops when $\sum_i F_i^2 \le 10^{-20}$. On this system that threshold is *below the arithmetic floor of the period map.* The capital and bond Euler equations difference $O(1)$ expectations down to $O(10^{-4})$

quantities, so each residual carries roughly 10^{-10} of unavoidable cancellation noise; no Newton step improves on it. Left alone, the solver burns its entire iteration budget backtracking on a converged answer.

The repair distinguishes two reasons the line search can fail:

$$\text{no admissible step} \quad \Longrightarrow \quad \begin{cases} \textbf{converged}, & \text{if } \max_i |F_i| \le \text{floor_tol}, \\ \textbf{failed}, & \text{otherwise.} \end{cases} \tag{32}$$

Reaching the floor *is* convergence; stalling at $\max|F| = 10^{-3}$ is a failure. The distinction has to be drawn explicitly, because the symptom — the line search cutting the step to nothing — is identical.

Remark 6.2 (Choosing the acceptance threshold on economic grounds). The shipped acceptance is $\max_i |F_i| \le 10^{-9}$. The justification is not numerical convention but a signal-to-noise argument: the headline risk shock moves the constraint multiplier by 7.5×10^{-3} and the labour residual by $\approx 7 \times 10^{-4}$, so 10^{-9} is four orders of magnitude below any economic signal in the model, while sitting one order above the $\approx 10^{-10}$ arithmetic floor. *Acceptance thresholds should be set by the size of the smallest effect one intends to report.*

6.4 Cost, and the one place parallelism is free

With $m = R \cdot 2n$ unknowns, each Jacobian costs $m + 1$ residual evaluations, each of which walks the period map at all $2n$ (point, regime) pairs. Hence

$$\text{cost per Newton step} \;=\; (m+1) \times 2n \times (\text{one period map}) \;=\; O\big(n^2\big), \tag{33}$$

quadratic in grid size. At the shipped 115 points, $m = 20 \times 2 \times 115 = 4600$, so one Jacobian is 4601 residual evaluations — about 25 minutes serially.

The columns of (30) are *independent by construction*, which makes the Jacobian the ideal parallel target: distribute columns across worker processes, each holding its own copy of F. Because every column is computed with the identical serial formula, the assembled matrix is *bit-identical* to the serial one — a property worth insisting on, since a parallel implementation that merely agrees to tolerance makes every subsequent debugging session harder.

Remark 6.3 (An operational hazard worth knowing). The workers are spawned per solve and import the model from disk at that moment. Editing model code while a run is in progress hands the workers *different code* from the main process. The result is a Jacobian that is internally inconsistent, with no error message.

6.5 Why Jacobian-free Newton–Krylov fails on this system

The $O(n^2)$ cost makes the Jacobian-free alternative tempting. Newton–Krylov solves $J\,\delta = -F$ by GMRES using only directional derivatives $Jv \approx [F(\boldsymbol{\theta} + hv) - F(\boldsymbol{\theta})]/h$, never forming J. It is implemented, tested, and *is not the default*, because it does not work here. From a warm start at $\max|F| = 1.9 \times 10^{-3}$, 25 outer iterations move the norm to 1.88×10^{-3} — nowhere.

The diagnosis is instructive, and it generalises. Krylov methods converge at a rate governed by the *conditioning* and spectral clustering of J. Here:

- the residual *rows* span four orders of magnitude — capital and bond Euler equations are in return units ($\sim 10^{-4}$), the identity residuals (28) are log ratios ($\sim 10^{-2}$);
- the *columns* are log-policies of very different scales.

Unpreconditioned GMRES on a badly scaled operator returns a search direction whose Armijo line search then cuts to nothing. A dense *LU* factorisation of the same matrix is indifferent to scaling in a way an iterative method is not, and it converges. *The lesson is that Jacobian-free methods trade memory for a dependence on conditioning, and in equilibrium systems with heterogeneous residual units that trade is often bad.*

6.6 Chord and Shamanskii variants

One can reuse a factorised Jacobian for several steps ("`jac_every`> 1"), trading quadratic convergence for cheaper steps — the Shamanskii iteration, with order $\nu + 1$ for ν reuses. Measured on the shipped grid, reuse went $1.9 \times 10^{-3} \to 1.8 \times 10^{-4} \to 1.1 \times 10^{-4}$, with the reused step barely moving, whereas a fresh Jacobian each step converges quadratically. Since the Jacobian dominates total cost, *fewer better steps beat more cheaper ones here*; the lever is left at 1 and kept available for a grid where a fresh Jacobian is genuinely unaffordable.

7 Homotopy, continuation, and warm starts

Newton's locality means that in practice *the solver is the ladder, not the Newton step.* This section is about that ladder, which is where most of the practical difficulty of global solution work actually lives.

7.1 The principle

Embed the problem of interest in a family indexed by a scalar $\varsigma \in [0, 1]$:

$$F(\boldsymbol{\theta}; \varsigma) = \mathbf{0}, \qquad \varsigma = 0 \text{ easy}, \qquad \varsigma = 1 \text{ wanted}, \tag{34}$$

such that the solution path $\boldsymbol{\theta}^\star(\varsigma)$ is continuous. Solve at $\varsigma = 0$, then step ς up, each time using the previous solution as the starting point. If the steps are small enough, each start lies inside the Newton basin of the next problem. Under regularity of the path this is the standard *predictor–corrector continuation* method; here the predictor is simply the previous solution (a zeroth-order predictor), which is adequate because the steps are small.

7.2 The four continuations in this solver

1. **Risk continuation.** Solve the no-default regime at $\pi = 0$ first. At $\pi = 0$ the default branch carries zero weight and the model is its risk-free counterpart — a much easier problem, and one whose solution is a good start for the risky one.

2. **Haircut homotopy.** The default regime is warm-started from the no-default solution and the recovery rate is walked *down*: $0.85 \to 0.70 \to 0.55 \to 0.45$. Each rung is a genuine root. A 55% write-down solved cold is infeasible; solved as the last rung of this path it is routine. This is continuation in a *structural* parameter and it is the technique that makes large feared events tractable.

3. **Policy-cap homotopy.** The central-bank backstop caps a sovereign spread. The cap is walked down from a level at which it binds nowhere (700 bp) to the target (200 bp) in 50 bp rungs, with a failed rung retried at half the step down to a minimum. This is adaptive step control: the step size responds to observed failure. Note the economics of why it is needed — a purchase moves the bond price only through the continuation and by very little, so a cap far below the market spread is met only at a *corner* (the central bank holding the bank's entire book), and the Newton reaches a corner only by degrees.

4. **Smoothing continuation.** The complementarity smoothing η is itself walked in, $10^{-2} \to 10^{-3} \to 10^{-4}$, on the refined grid. A wide smoothing makes the regime switch a gentle transition the Newton can follow; each tighter stage starts next to its own answer. With the final smoothing imposed from the start, the refined solve stalled with steps cut to $1/256$.

5. **Grid refinement as continuation.** The dense factor in the risk dimension is walked up ($3 \to 5 \to 9$ nodes), each grid seeded by evaluating the previous solution at the new nodes. Because both grids are drawn on the *same* box this is interpolation, not extrapolation. Going straight from 3 nodes to 9 asks the Newton to start from a seed that is badly wrong at the new interior nodes — the coarse quadratic reads $p^d = 1.82\%$ where the truth is 0.67%.

7.3 Why the warm start is not optional, quantitatively

It is tempting to treat warm starts as a convenience. The measurements say otherwise. Seeding the refined grid directly from the coarse solution leaves $\max|F| = 4.0 \times 10^{-2}$, which is outside the Newton basin; 20 sweeps of the (slow, globally stable) time iteration bring it to 1.9×10^{-3}, which is inside. The reason is structural: the refined grid visits combinations — high deposit obligation *and* high risk — for which the coarse interpolant has *no cross term at all*, by Proposition 3.1. Its value there is not merely inaccurate, it is an extrapolation of a basis that does not contain the relevant interaction.

This is the role in which time iteration survives: not as a convergent solver, but as a *basin-finder* for the Newton. The division of labour is exact — the globally stable but slowly converging method delivers a point inside the basin, and the locally convergent but globally unreliable method finishes the job to the floor.

> **In the code.** `solve_recursive` in `recursive_experiment.py` is the ladder; `_stage` wraps "short time-iteration warm start, then global Newton". The rungs are `_solve_baseline` (risk + haircut), `_solve_tpi` (cap homotopy with half-step retry), `_refine_s` (grid ladder with the smoothing ladder nested inside), and `_seed_from` does the interpolation between grids. `_stamp_verdict` records on the rule set itself whether every rung actually rooted — without which a reader cannot distinguish a converged solve from a stopped one.

Part IV Nonsmoothness: Constraints and Expectations

8 Complementarity: writing an occasionally binding constraint

8.1 The KKT system and its closed form

The intermediary's constrained problem yields the complementarity system

$$\mu \geq 0, \qquad \text{slack} \equiv \alpha n - \lambda A \geq 0, \qquad \mu \cdot \text{slack} = 0, \tag{35}$$

together with the asset Euler equations. Solving (35) as an inequality-constrained system at every grid point would be expensive and fragile. The model's structure supplies something better: because the banker's objective is linear in the portfolio, the multiplier has a *closed form*. Combining the balance-sheet identity with the Euler equations gives

$$\boxed{\mu(\mathbf{S}) \;=\; \max\left\{ 1 - \frac{\mathbb{E}[\Omega']\,(1 + r^{dep})\,n}{\lambda\, A} \;,\; 0 \right\}} \tag{36}$$

which is explicit, bounded in $[0, 1)$, and automatically satisfies (35). The remaining Euler equation — $\mathbb{E}[\Omega'(R_K - R)] = \lambda_K \mu$ — is then a plain residual pinning next-period capital, *valid in both regimes*. The occasionally binding constraint has been reduced to one max operator and one smooth equation.

This is a general and underused lesson: before discretising an inequality, check whether the model's own first-order conditions solve for the multiplier analytically. Linearity in the choice variable is often enough.

8.2 Three ways to write a complementarity residual, and why the choice matters

Where a closed form is *not* available — for instance the corner condition on bank holdings when the central bank may absorb the entire book — the complementarity must enter as a residual. There are three standard choices, and they are not interchangeable.

(i) The max **form.** Write the residual as $\mu - \max\{\text{expr}, 0\}$. Exact, but C^0: the finite-difference Jacobian straddling the kink is wrong, and the fitted rule inherits the Gibbs phenomenon of Figure 2.

(ii) The min **form.** Write $\min(a, b) = 0$ with $a = \mu$, $b = \text{slack}$. This is correct as a mathematical statement and *badly behaved as a residual*. The defect is visible in Figure 2(c): whenever $b < a$, $\min(a, b) = b$ has *zero slope in* a. In the backstop application this meant the residual had no slope in the purchase quantity while the price sat below the floor, so the Newton could only move the price — and stalled.

(iii) The Fischer–Burmeister form. Define

$$\text{FB}_\eta(a, b) \;=\; a + b - \sqrt{a^2 + b^2 + \eta^2}. \tag{37}$$

Proposition 8.1 (Properties of (37))**.** *At $\eta = 0$, $\text{FB}_0(a, b) = 0$ if and only if $a \geq 0$, $b \geq 0$ and $ab = 0$: the function's zero set is exactly the complementarity set. For $\eta > 0$ the zero set is $\{a, b > 0 : \; ab = \eta^2/2\}$, a hyperbola approaching the complementarity corner as $\eta \to 0$. The partial derivatives are*

$$\frac{\partial \text{FB}_\eta}{\partial a} = 1 - \frac{a}{\sqrt{a^2 + b^2 + \eta^2}}, \qquad \textit{and symmetrically in } b, \tag{38}$$

which are strictly positive in both arguments *for $\eta > 0$, including when one argument sits at zero. The zero set is invariant to common scaling of (a, b, η).*

Retaining slope in both arguments is precisely what the min form loses, and it is why (37) is the right residual for a Newton method. The scaling invariance is also practically useful: it means one may normalise a and b by steady-state magnitudes without changing the solution.

8.3 Smoothing: how wide, and against what

Smoothing replaces a kink by a C^∞ approximation. For max:

$$\text{smax}_\eta(x, c) \;=\; c + \tfrac{1}{2}\Big[(x - c) + \sqrt{(x - c)^2 + \eta^2}\Big]. \tag{39}$$

Proposition 8.2 (Smoothing bias)**.** *For $g = x - c$ with $|g| \gg \eta$,*

$$\text{smax}_\eta(x, c) - \max(x, c) \;=\; \frac{\eta^2}{4\,|g|} + O\big(\eta^4/|g|^3\big), \tag{40}$$

so the bias is second order *in the smoothing width and inversely proportional to the distance from the kink. At the kink itself ($g = 0$) the bias is $\eta/2$.*

Proof. For $g > 0$, $\text{smax} - \max = \frac{1}{2}\left[\sqrt{g^2+\eta^2} - g\right] = \frac{g}{2}\left[\sqrt{1+\eta^2/g^2} - 1\right] = \frac{g}{2}\left[\frac{\eta^2}{2g^2} + O(\eta^4/g^4)\right]$. The case $g < 0$ is symmetric. □

Proposition 8.2 is what makes smoothing widths choosable rather than arbitrary. Two constraints bracket η from both sides:

From above: the bias must be economically invisible where the model spends its time. At $\eta = 10^{-5}$ and the measured gaps, the bias at the steady-state rest point is $\approx 10^{-11}$ — three orders below the solver's acceptance.

From below: η must exceed the finite-difference step ε of (30), or the Jacobian straddles the (nearly) kinked region and is simply wrong. This is what forced the backstop smoothing *up* from 10^{-5} to 10^{-4}: at 10^{-5}, against $\varepsilon = 10^{-6}$, the Jacobian near the floor's boundary was off by several percent and the refined solve converged only linearly ($7.6 \to 4.5 \to 3.2 \times 10^{-4}$).

So $\varepsilon \ll \eta \ll$ (economic scale), and the shipped pair $(\varepsilon, \eta) = (10^{-6}, 10^{-4})$ sits in that window. *Smoothing width and differencing step are a joint choice; tuning one without the other is how a Newton method silently loses its quadratic rate.*

Remark 8.3 (Smoothing is not only for Newton: it protects the fit). A *hard* clip applied to a value that is subsequently *fitted* puts a kink inside the collocation box. One saturated node was measured to move the interpolant at the ergodic centre by 26% and could drive it negative. Similarly, a hard floor is a *plateau with zero gradient*: an iterate that lands on it makes the finite-difference Jacobian see nothing, and the point sticks forever (one documented case sat at $|F| \sim 4 \times 10^8$ indefinitely). Every guard in the period map is therefore smoothed, with the single exception of the KKT switch in (36) — which is left hard, because it *is* the economics rather than a numerical guard.

8.4 Garcia–Zangwill: one smooth variable for two branches

A particularly elegant device deserves its own treatment. Consider a policy with two mutually exclusive branches: either the central bank buys ($m > 0$) and the price sits exactly at the floor, or it does not buy ($m = 0$) and the price sits above the floor. The complementarity is $m \geq 0$, $Q - Q^{\text{floor}} \geq 0$, $m\,(Q - Q^{\text{floor}}) = 0$.

The Garcia–Zangwill reformulation introduces a *single* unknown x carrying both branches:

$$\boxed{m = B \cdot \max(x, 0), \qquad Q = Q^{\text{floor}} + Q^{ss} \cdot \max(-x, 0).} \tag{41}$$

Complementarity now holds *by construction* — at most one of the two max terms is non-zero — and, crucially, the *stored rule* $x(\mathbf{S})$ is smooth across the switching boundary even though $m(\mathbf{S})$ and $Q(\mathbf{S})$ are not.

That last property is the point. Storing m directly carries the kink into the Chebyshev fit, with the pathologies of Figure 2: on the coarse grid the quadratic through $(0, 0, 0.29)$ bought 19.7% of the bond stock at a shock where the floor does not bind, and a degree-4 fit turned the same kink into *sales* between nodes. Fitting the smooth x and recovering the kinked objects pointwise through (41) eliminates both artefacts.

8.5 Root in one form, read in the other

The final refinement is a trick with a general moral. Empirically:

- the Garcia–Zangwill form (41) is the right one to *store and read*, since its rule is smooth;
- the Fischer–Burmeister form (37) is the right one to *root*, since the Newton stalled walking the policy cap down in GZ form (steps cut to 1/64) where FB takes full steps.

Both forms describe the *same allocation at the collocation nodes*, so the conversion between them is exact there (measured: the GZ residual evaluated at the FB solution is 2.2×10^{-9}). The solver therefore roots in FB, converts at the nodes, and re-roots in GZ in one or two polishing steps. *A reformulation that is good for the solver and a reformulation that is good for the approximation need not be the same reformulation, and when they agree at the nodes one may use both.*

In the code. `_fb`, `_smax`, `_smin`, `_sclip` in `point_map.py` are (37) and (39); `_tpi_purchase_and_price` is (41); `_incentive_constraint` is (36) with the hard KKT max and a smoothed upper guard. The form switch is `_tpi_form`/`_tpi_polish` in `recursive_experiment.py`, selected by `cal["tpi_gz"]`.

8.6 The kink is also an accuracy limit, and it must be reported

The preceding devices make the *solver* work. They do not make the *approximation* exact, and in this model the residual inexactness is economically material. The argument is worth following closely, because it generalises to any model with an ergodically binding constraint.

The multiplier (36) is C^0, and at this calibration $\mu = 0$ *exactly* at the stochastic rest point: the economy sits *on* the kink. By Figure 2(a), a Chebyshev interpolant of a max returns strictly positive values in a neighbourhood where the truth is zero. Therefore:

- **reading the fitted rules** at the rest point prices a credit spread that is not there, and output falls;
- **clearing the period map exactly** at the same state keeps $\mu = 0$ until $p^d \approx 1.5\%$, and output *rises* (the deposit rate falls with no spread to offset it).

Measured at $p^d = 1.98\%$: output -0.081% fitted against -0.008% cleared. *The two reads disagree by more than the response being measured.* Refinement helps slowly ($0.087 \to 0.073$ percentage points going from 21 to 95 points), exactly as a Gibbs phenomenon does.

Three responses are available, and the repository uses all three.

1. Report the pair, not a point. Both reads are computed and printed side by side at every state. Neither is "the truth"; the honest object is the bracket. It is worth noting that this pathology is not peculiar to this model: the same measurement applied to Bocola's own published solution gives a fitted liquidity premium of 28.4 bp on impact where the exact multiplier gives 2.1 bp — and 28.4 is the published number.

2. Move the model off the kink by calibration. The gap is largest when the constraint is exactly marginal. Raising the steady-state credit-spread target lifts the ergodic multiplier off zero and collapses the bracket:

spread target	μ_{rest}	rest spread	Y fitted	Y exact		|gap| (pp)
8 bp/yr	0.00000	0.0 bp	-0.094%	-0.007%	0.087	(on the kink)
25 bp/yr	0.00000	0.0 bp	-0.097%	-0.061%		0.036
100 bp/yr	0.00983	79.0 bp	-0.102%	-0.123%	**0.021**	(shipped)
250 bp/yr	0.02860	228.7 bp	-0.096%	-0.118%	0.022	(no further gain)

A fourfold collapse of the identification gap at 100 bp, with nothing more to gain beyond. Note what does *not* work: varying the banker exit rate f. The bank calibration forces $\alpha^{ss} = \lambda\theta$ at the steady state whatever f is, so the binding margin is $\approx \theta\mu^{ss}$ and μ^{ss} is set by the *spread*, not by f. Identifying which parameter actually moves an approximation pathology requires tracing the calibration map, not guessing.

3. State the limit. At this resolution the *level* of the output response is bracketed, not identified. The sign, the mechanism, and the relative magnitudes are robust; the second significant figure of the level is not. That is a legitimate result to report, provided it is reported.

9 Expectations: quadrature over a mixed shock

Every residual contains expectations over two objects of different type: a continuous innovation to the risk factor, and a discrete default realisation.

9.1 Gauss–Hermite for the continuous part

For $\epsilon \sim N(0,1)$ and a smooth integrand, Gaussian quadrature chooses both nodes and weights optimally:

$$\mathbb{E}\big[h(\epsilon)\big] \;\approx\; \sum_{q=1}^{n_{gh}} w_q \, h(\epsilon_q), \tag{42}$$

exact for all polynomials of degree $\leq 2n_{gh} - 1$ — twice the accuracy of a rule with fixed nodes, which is the whole point of Gaussian rules. With $n_{gh} = 5$ the rule is exact to degree 9, ample for the smooth parts of the integrand.

Remark 9.1 (The $\sqrt{2}$ trap, and why it is worth a paragraph). Two conventions for Hermite quadrature circulate and they differ by $\sqrt{2}$:

$$\textit{probabilists'}: \quad \int h(x)\, e^{-x^2/2}\, dx \quad \Rightarrow \quad \text{nodes already in standard-deviation units}, \tag{43}$$

$$\textit{physicists'}: \quad \int h(x)\, e^{-x^2}\, dx \quad \Rightarrow \quad \text{nodes in units of } \sigma/\sqrt{2}. \tag{44}$$

Applying the map $s' = \text{mean} + \sigma\epsilon_q$ with *physicists'* nodes silently rescales the innovation by $1/\sqrt{2}$. The consequence is not academic: a reference model's reported innovation standard deviation of 0.63 corresponds,

inside its own solved solution, to an effective $0.63/\sqrt{2} = 0.4455$. Feeding 0.63 into a correct probabilists' rule reproduces the published *parameter* but not the published *model*. The measured difference is substantial — unconditional standard deviation of s of 2.02 versus 1.43, ergodic mean default probability 0.66% versus 0.27%, bond marked at 0.9057 versus 0.9178, steady-state credit spread 32.8 versus 25.5 bp. *When replicating, check the quadrature convention before concluding anything about a parameter.*

9.2 The product measure over the discrete regime

The default indicator $d' \in \{0, 1\}$ is discrete with priced probability $p^d(s)$, and the two regimes have *separate* decision rules. The joint measure is a product:

$$\mathbb{E}\big[h\big] \;=\; \sum_{q=1}^{n_{gh}} \sum_{d' \in \{0,1\}} \underbrace{w_q \cdot \big[\, d'\, p^d(s) + (1 - d')(1 - p^d(s)) \big]}_{\text{cell weight}}\; h\big(\epsilon_q,\, d'\big), \tag{45}$$

with cell weights summing to w_q in each ϵ-column, so the quadrature measure is preserved exactly. Note what (45) is *not*: it is not a representative branch, not a certainty-equivalent, and not an expected-loss add-on to a risk-free price. The default state is evaluated as the *same fitted decision rules at a reachable next-period point*, so the risk premium is endogenous — $\Omega^d > \Omega^{nd}$ because marginal utility is high on low payoffs.

This matters because the alternative was tried and gets the sign wrong. A perfect-foresight representative-branch treatment of the same mechanism made sovereign risk *expansionary*. The economics requires the covariance, and the covariance requires genuine multi-branch integration.

9.3 Exact nesting at zero risk

A property worth engineering for deliberately: at $p^d \equiv 0$ the default regime carries zero weight, and the implementation *skips* it and aliases it to regime 0. The consequence is that $\pi = 0$ nests the risk-free model *exactly* — bit-for-bit — rather than to solver tolerance. This gives a sharp regression test ("switch risk off, recover the known model"), which is far more informative than a tolerance comparison, and it saves a full grid interpolation plus two capital blocks per node.

Remark 9.2 (Quadrature order must travel with the solution). A solution computed at $n_{gh} = 5$ and *read back* at $n_{gh} = 7$ differs, and the difference would be charged to whatever economic question is being asked. The rule set therefore records the order it was solved under, and every reader — impulse responses, decompositions, accuracy reports — takes the order from that stamp rather than from a default.

> **In the code.** `gh_nodes(n)` uses `numpy.polynomial.hermite_e.hermegauss` — the probabilists' rule, deliberately. `_regime_weights` is the cell weight in (45); `_expect` is the double sum; `_continuation` performs the aliasing at zero weight. The stamp is `RuleSet.n_gh`.

Part V The Model-Side Levers

10 Designing the state vector

The state vector is the most consequential single choice in a global solution, because every cost in Part II scales in its length and every accuracy claim is conditional on it. The discipline is: *a variable is a state if and only if some period-t decision depends on it and it is not recoverable from the others.*

This model carries $d = 12$:

$$\mathbf{S} = \big[\, K_D,\, K_F,\, P_D,\, P_F,\, b_{DD},\, b_{DF},\, b_{FD},\, V,\, s,\, Z_D,\, M,\, O \,\big]. \tag{46}$$

States	Object	Why it must be remembered
K_D, K_F	capital stocks	predetermined: the stock producing at t was bought at $t-1$
P_D, P_F	gross deposit obligations	the rate paid at t was locked at $t-1$
b_{DD}, b_{DF}, b_{FD}	sovereign holdings	the perpetuity pays *per holder*, so last period's *split* is payoff-relevant once each bank bids through its own FOC
V	cross-border deposit position	see §10.1
s	risk factor	the exogenous driving process; AR(1)
Z_D	TFP	deterministic AR(1); exercised only off the ergodic set
M, O	central-bank book	O carries *last period's price*, which nothing else remembers

Note the recurring logic: *every* state here exists because some payoff or constraint at t is a function of a $t-1$ decision. Three of them became states only when the corresponding market was made to clear — while a bond's split was a fixed steady-state share, it carried no information; once each bank bids through its own first-order condition, the split is genuinely part of the state. The modelling choice and the state-space choice are the same choice.

10.1 Two case studies in state choice

These are worth reading carefully, because they illustrate a general principle about *error propagation* through a state vector that is rarely made explicit.

Case 1: carry the small difference, not the large level. Under national deposit clearing, each household's gross claim coincides with its own bank's obligation, and seven states suffice. Under a *union* deposit market the two diverge, and the divergence must be remembered. Two parameterisations are algebraically equivalent:

$$\text{(a) carry } W_D: \qquad W_F = (P_D + p\,P_F - W_D)/p, \tag{47}$$

$$\text{(b) carry } V = W_D - P_D: \qquad W_D = P_D + V, \quad W_F = P_F - V/p. \tag{48}$$

Option (a) was measurably worse. W_F inherits the fit error of P_D, P_F *and* W_D simultaneously, and because consumption is a small difference of large terms ($C_F \approx 0.79$ from gross legs of size ≈ 8), a 0.06% error in W_F became a 1.1% error in the F Euler residual — 22 times the D-side error, with correlation 0.97 between the two. Option (b) carries a quantity that is *zero at the symmetric steady state and small everywhere*, so neither claim is a difference of large numbers and the two countries are symmetric in their error.

> *Principle.* Among algebraically equivalent state vectors, prefer the one whose components are small and whose reconstructions are not differences of large numbers. Approximation error does not respect algebraic equivalence.

Case 2: states that are zero at the steady state nest exactly. The backstop's two states M and O are zero at the steady state with bands symmetric around zero. Hence the steady state remains a collocation node, *and* every node with $M = O = 0$ is a node of the smaller grid. On that slice the 12-state interpolant *is* the 10-state one, so the policy-off model nests the no-policy model exactly rather than approximately. Designing a new state to be zero at the reference point is cheap and buys an exact nesting test.

11 The parametric levers

This section catalogues the parameters that move the *character* of the solution, as opposed to its calibration targets. The division is not sharp, which is itself the point: in a global solution, several parameters that look purely economic are also numerical levers.

11.1 Preferences, technology, and why GHH changes the algorithm

The preference specification is GHH — utility over the composite $x = C - v(N)$ — rather than separable. This is a *mechanism* choice with a numerical consequence worth drawing out. GHH removes the labour-supply wealth effect, which in the closed-economy reference model is the dominant transmission channel ($d\log\ell = -1.25\,d\log c$ exactly). With GHH, output moves *only* through the working-capital wedge

$$r^{wc} = r^{dep} + \frac{\lambda_K\,\mu}{\mathbb{E}[\Omega]}, \tag{49}$$

so the solution's entire real response is routed through the multiplier μ — the one object in the model that is non-differentiable. *The choice of preferences determines how much of the answer depends on the approximation's behaviour at the kink.* That is why §8.6 is not a footnote in this model.

A second consequence: the composite x must be floored smoothly, not clipped. A hard floor at 10^{-9} against $x^{ss} = 0.391$ is eight orders away and yet was reached by a corner iterate, whereupon the zero-gradient plateau froze the point permanently. The floor is instead set at 5% of steady-state consumption and smoothed — inactive anywhere near the ergodic set, where $x/C = 0.50$, but with a usable derivative if an iterate strays.

11.2 The bank block: a scalar fixed point with a fold

The bank calibration is the most mathematically interesting part of the parameterisation, because it is a *fixed-point problem that can fail to have the root one wants.*

The franchise value satisfies

$$\alpha = \frac{\Omega\,(1 + r^{dep})}{1 - \mu}, \qquad \Omega = \beta\big[f + (1-f)\alpha\big], \qquad \mu = \frac{\Omega\,s\,\theta}{\alpha}, \tag{50}$$

where s is the target credit spread, θ the target leverage, and f the banker exit rate. Substituting makes (50) a scalar equation in α. It generically has *two* roots.

Proposition 11.1 (Why the least root). *The map $\alpha \mapsto \Omega(\alpha)(1 + r^{dep})/(1 - \mu(\alpha))$ has slope $\beta(1 - f)(1 + r^{dep})/(1 - \mu)$, which exceeds 1 once $\mu \gtrsim 0.04$. The fixed-point problem therefore* folds*: below the fold there is a stable low root, above it an unstable high root, and for extreme targets no root at all. Value iteration from below converges to the* least *root; the calibration must select the same one, or the model and its calibration describe different economies.*

The implementation therefore does not hand (50) to a root-finder with a guess. It performs a *sign-change scan on a logarithmically spaced grid* over $[10^{-6}, 10^{6}]$ and brackets the *first* crossing, then refines with Brent's method. This is slower than a Newton step and is the right choice: a scan cannot land on the wrong root, and the failure mode it prevents (silent selection of the unstable branch) produces a model that solves cleanly and is wrong.

Remark 11.2 (The fold as a feasibility boundary). Targets can be jointly infeasible. At the shipped exit rate, a 400 bp spread target has no root — the fold has swallowed it. The steady-state solver asserts on the bank identity rather than warning, with an error message naming the fold, because the alternative is a converged solve with a negative multiplier.

Having selected α, the two frictions are recovered in closed form: $\lambda = \alpha/\theta$ and $\omega_{\text{ent}} = D/\theta - (1 - f)s$ with $D = 1 - (1 - f)(1 + r^{dep})$, with an explicit feasibility check $\omega_{\text{ent}} > 0$.

11.3 A worked identification exercise: which lever moves what

The following is a model of how to diagnose a lever, and the answer is counter-intuitive. Suppose one wants the *ergodic* credit spread to hit a target. The obvious instrument is the *steady-state* spread target. Measured:

Instrument moved	Response of the ergodic spread	Verdict
spread target, 8.04 → 0.04 bp	32.8 → 24.6 bp, λ fixed at 0.19998	**dead lever**
leverage target	sets $\lambda = \alpha/\theta$ directly	live
exit rate f	sets the fold, hence α	live

The spread target is nearly inert because $\lambda = \alpha/\theta$ and, as $\mu \to 0$, $\alpha \to \Omega(1 + r^{dep})$: λ is pinned by the *leverage* target. The spread enters only through μ inside the fold, which is second order at small μ — a flat map, on which a secant iteration stalls. *Before iterating on an instrument, differentiate the calibration map; a flat map wastes hours and produces a plausible-looking failure.*

11.4 The risk process and the duration interaction

One relationship here is frequently misstated and is worth getting right, because it is a genuine interaction rather than a single-parameter effect.

> **Duration alone does not generate mark-to-market losses. Duration *interacting with* persistence does.**

Holding duration fixed at the calibrated 17 quarters and varying ρ_s, a perpetuity and a one-quarter bill lose *exactly the same* on the reported impulse when $\rho_s = 0$ (both -1.01%), diverge mildly at $\rho_s = 0.5$ (-1.27% vs. -1.01%), and separate only at the calibrated $\rho_s = 0.95$ (-5.58% vs. -1.01%, a factor of 5.5). The reason is elementary once stated: a *transitory* risk shock marks a long bond down no more than a short one, because the hazard is elevated only briefly. The loss is large because a *persistent* s_t keeps the hazard elevated across the claim's whole remaining life. Two corollaries: a short bond's *principal* is marked down too, so the distinction is never coupon-versus-principal; and any statement about the mechanism must carry the persistence, not just the maturity.

11.5 Fiscal feedback: calibrate the root, not the coefficient

The government follows a Bohn-type rule, taxes responding to the debt level. The natural parameterisation is the response coefficient γ_τ. The implementation instead calibrates the *debt root* — the autoregressive coefficient of the debt stock the rule delivers — and inverts for γ_τ:

$$\text{root} = (1 - \delta_b) + \frac{\delta_b - \gamma_\tau}{Q_B^{ss}} \qquad \Longrightarrow \qquad \gamma_\tau = \delta_b - \big(\text{root} - (1 - \delta_b)\big)\, Q_B^{ss}. \tag{51}$$

Three reasons, each generalisable:

(i) The root is the object with economic content. A coefficient of 1.0 sounds disciplined and, on this calibration, makes a 55% haircut a 53%-of-GDP tax *windfall*, leaving the sovereign more indebted after default than before.

(ii) Units are treacherous. Read as an *elasticity* instead of a level coefficient, the same "$\gamma_\tau = 1$" does not stabilise debt at all here, because steady-state taxes are 0.00297 against debt of 0.98: $dT/dB = 0.003$

and the root is 1.0002. A rule imported from a model where taxes are a substantial share of GDP does not transfer to one where they are 0.3% of it.

(iii) The root and the box are a joint choice. Peak debt deviation after an impulse scales as impulse/(1 − root). At an earlier root of 0.9929 (half-life 97 quarters) the debt stock marched past +20% and hit the collocation box wall at quarter seven — and the resulting figure's "trough then flat recovery" *was that wall*, not economics. At the shipped root of 0.93 (half-life 9.5 quarters) peak deviation is near +6%, inside the band the grid carries.

> *Principle.* Parameterise a policy rule by the *persistence it delivers*, then check that persistence against the width of the approximation domain. A dynamic parameter and a grid band are one decision, not two.

Note that (51) leaves the steady state untouched for any root, since at the anchor the rule returns steady-state taxes identically. No bank or household recalibration follows a change of root — a desirable property to engineer into a rule.

11.6 Inducing stationarity: the SGU premium

In an open economy with incomplete markets the net foreign asset position typically has a unit root, and a global solver cannot live with one: the simulated path wanders out of the box. The standard repair (Schmitt-Grohé–Uribe) is a small debt-elastic premium on the external position, here entered in the deposit parity condition with coefficient $\kappa_{nfa} = 0.01$.

The lever is undistorting at the symmetric steady state (where the position is zero) and induces stationarity only off it, so it costs no steady-state fidelity. What it buys is not cosmetic: at $\kappa_{nfa} = 0$ the model has no force returning wealth to the symmetric steady state, and 24 quarters after a shock that had 88% decayed, output was still −0.373% *and falling*, with every stock permanently displaced. *Reading a trough off a non-stationary path is meaningless*; the premium's job is to make the impulse response a well-defined object at all.

11.7 Guards: parameters that exist only to keep the Newton alive

These have no economic content and must be documented as such, with evidence that they are inactive where it matters.

Remark 11.3 (The discipline that makes guards acceptable). An exception raised inside one finite-difference column kills the entire Jacobian — so the period map must be *evaluable everywhere in the box*, including at states the model will never visit. That is the honest reason guards exist. The corresponding discipline is: every guard is smoothed, every guard's inactive margin is measured and recorded, and every guard is exposed as a parameter so its effect can be tested rather than trusted. A guard whose bite has never been measured is an unlabelled modelling assumption.

11.8 Asymmetric country size and a derived parameter

The two countries differ in mass, $\text{size}_F/\text{size}_D = 8$, while the *per capita* steady state is identical in both. The mass ratio enters only where D and F quantities are aggregated. Two features are worth noting as technique.

First, home bias *must* scale with size or trade cannot balance. At $p = 1$ balanced trade requires $\text{size}_D(1-\omega_D) = \text{size}_F(1-\omega_F)$, so

$$(1-\omega_F) = (1-\omega_D)\cdot\frac{\text{size}_D}{\text{size}_F} \tag{52}$$

is *derived* in the calibration rather than set. Deriving a parameter that is implied by an identity, rather than entering it independently, removes a whole class of silent inconsistency — the two can never fall out of step.

Second, the reason for the asymmetry is numerical-economic rather than descriptive. With a symmetric union, D is half the union, so D's own sovereign shock moved the union real deposit rate by 45 bp/yr and cancelled 78% of the credit-spread rise *before it reached any firm's wage bill* — through the second leg of (49). Setting $\text{size}_F = \text{size}_D$ nests the symmetric model exactly, so the asymmetry is a testable switch rather than a rebuild.

12 Transforms, scaling, and conditioning

The final class of levers concerns the *coordinates in which the unknowns and residuals are expressed.* These are invisible in the economics and decisive in the numerics.

12.1 Fit in logs, store in levels

Policies that must remain positive are *fitted* in logs:

$$\mathcal{T}(x) = \log x \quad \text{(quantities, prices)}, \qquad \mathcal{T}(r) = \log(1+r) \quad \text{(rates)}, \tag{53}$$

with values stored in levels throughout so that warm starts, damping and every caller are unchanged. Only the Chebyshev fit and evaluation pass through the transform.

Three benefits, in increasing subtlety:

1. **Positivity by construction.** exp of any polynomial is positive, so the hard clips that previously protected positivity become unnecessary — and by the remark in §8.3, removing a clip removes a kink from inside the box.
2. **The fit no longer has to represent a plateau.** A clipped policy is constant over a region; a polynomial fitting that region wastes degrees of freedom and oscillates at its edge.
3. **Rates that may go negative are handled by the *gross* transform.** $\log(1+r)$ is defined for $r > -1$, so a legitimately negative rate is representable while the positivity of the gross rate is enforced.

A clamp of ± 50 on the exponent is retained purely to keep a diverging transient iterate finite — exp overflows near 709 and an infinity propagates into every expectation — and spans 10^{-22} to 5×10^{21}, unreachable by any admissible value.

12.2 Residual scaling is load-bearing

A Newton method applied to F and to DF for diagonal $D > 0$ has the *same* root and *different* numerical behaviour, because the line search compares $\|F\|_2$ across equations of different units. Every residual in this system is therefore normalised into $O(1)$ units:

Residual	Normalisation	Why that one
capital Euler	$\div\, \mathbb{E}[\Omega]$	leaves it in *return* units; dividing by $\lambda_K \mu^{ss}$ instead amplifies by $\approx 130\times$
labour FOC	$\div$ the real wage	relative wedge
goods market	$\div\, Y^{ss}$	share of output
deposit clearing	$\div$ steady-state deposits	share of the market
bond FOCs	$\div\, \mathbb{E}[\Omega]$	return units
identity residuals	log ratio	scale-free by construction ((28))

The choice of *which* normalisation is not free: as §6 noted, the surviving four-order spread across residual rows is exactly what defeats the Krylov backend. Residual scaling is the cheapest available preconditioner and should be chosen deliberately.

Remark 12.1 (Floating-point associativity can matter at a knife edge). One block in the steady-state solver carries an explicit instruction to preserve operand order, because the stage-1 solve runs to 10^{-12} and some downstream policy solves sit close enough to a Newton knife edge that a one-ulp reassociation flips a rung from converged to stalled. This is a real phenomenon in tightly converged nonlinear systems and an argument for recording *bit-identical* reproducibility (§6) as an explicit requirement rather than an accident.

Part VI Reading a Global Solution

13 The stochastic rest point

A solved rule set is not an answer; it is an object from which answers must be extracted carefully. This section is about the extraction, and it contains the single most common error in applied global solution work.

13.1 The deterministic steady state is not where the model rests

The deterministic steady state is an exact rest point of the period map at zero risk. Verified directly: solved at $\pi \equiv 0$, the model sits on it for 200 quarters to six decimal places with $\max|F| \sim 10^{-7}$. The steady state and the solver are both exact.

But the *risk-pricing* rules rest somewhere else. A risk-pricing economy does not rest where its risk-free counterpart does — precautionary motives, risk premia and the convexity of the constraint all displace the ergodic mean. Measured displacement at this calibration:

Y_D	-0.111%	K_D	-0.541%
C_D	-0.130%	b_{DD}	$+2.87\%$
I_D	$+0.246\%$	bank net worth	$+2.24\%$
μ_D: 0.0072 $\longrightarrow$ **exactly** 0 — the constraint is *slack* where the model lives			

This is not solver error, and three pieces of evidence establish that. It is *grid-independent* (coarse and refined grids agree to within 10% of the gap); it is a *unique global attractor* (eight perturbed starting points converge to the same state to 10^{-12}); and the reference model in the literature exhibits the same gap and locates it the same way.

13.2 Locating it, and differencing twice

The rest point is found by a zero-shock simulation to convergence: iterate $\mathbf{S} \leftarrow H(\mathbf{S}, g(\mathbf{S}))$ with the innovation set to zero until $\|\Delta\mathbf{S}\| < 10^{-11}$. Then *every* impulse response

1. **starts** at the rest point, not at the deterministic steady state; *and*
2. **is differenced** against an *unshocked path* from the same starting point:

$$\widehat{y}_t \;=\; \frac{y_t^{\text{shocked}}}{y_t^{\text{unshocked}}} - 1. \tag{54}$$

Both steps are necessary. Step 1 ensures the deviation is taken from the level the economy inhabits. Step 2 removes any residual drift along the unshocked path — including the approximation's own drift, which cancels because it afflicts both paths equally.

Remark 13.1 (What skipping this costs, measured). Starting at the deterministic steady state and differencing against a *fixed* base charges the walk between the two rest points to the shock. At quarter 12 the output response then reads +0.0651% where the correctly differenced response is +0.0300%: 54% *of the reported hump was drift*. Bank net worth read +4.20% against a true +1.55%, and an apparent "capital grinding down" was almost entirely drift. An impulse response computed this way is not approximately right; it is reporting a transition between two steady states as if it were a response to a shock.

13.3 Read the rules; do not re-solve

When evaluating at a state, one may either (a) read the fitted interpolant's own policy values, or (b) re-solve the period-map system at that state. These differ, and (a) is the correct default for the reason given in §6: a root-finder warm-started near a marginally binding constraint slips onto the slack branch, flipping the sign of the entire response. The re-solve path is retained in the code but switched *off*, with the measurement recorded — precisely because "the residual is large, so re-solve" is the obvious and wrong instinct.

The exception is the deliberate *bracket* of §8.6, where both reads are computed and reported as a pair.

14 Accuracy assessment

Convergence of the solver is necessary and far from sufficient: $\max|F| = 10^{-9}$ *at the collocation nodes* says nothing about the error *between* them. Four independent diagnostics are used, and the set is a reasonable template for any global solution.

14.1 Euler-equation errors on the ergodic set

The referee's question is: how wrong is the solution *where the model actually spends its time*? The answer is produced by simulating the solved rules forward for $\sim 10^3$ periods with drawn innovations and, at every visited state, evaluating all 20 equilibrium conditions *at the rule-implied policy* with no re-solving. The residual distribution is reported per equation in $\log_{10}$ units (mean, max, 90th percentile).

The units are worth dwelling on. A $\log_{10}$ residual of -4 in a consumption Euler equation means the household is making a consumption mistake of order 10^{-4} in relative terms — an interpretable economic magnitude, not an abstract tolerance. Reporting the *distribution* rather than a maximum matters because global methods characteristically have small errors in the interior and large ones at corners the model rarely visits.

14.2 Box escapes as the invariance check

The simulated path is deliberately *not* clipped before being recorded, and the fraction of periods spent outside the box is reported per state together with the worst overshoot as a fraction of box width. This is the empirical counterpart of the invariance analysis in §4: the theory says no axis-aligned box is one-step invariant, and this diagnostic says how much that matters in practice. A solution with small Euler errors and frequent box escapes is *extrapolating*, which the error metric alone would not reveal.

14.3 A convergence study, reported in the units of the question

Doubling resolution and reporting the movement is the only direct evidence that a solution is resolved. Going from 5 to 9 dense nodes in the risk dimension:

	$m = 5$ (115 pts)	$m = 9$ (207 pts)	move
impact output response (fitted)	−0.1105%	−0.1087%	1.6% of itself
impact output response (exact)	−0.1278%	−0.1234%	3.4% of itself
credit spread	+90.6 bp/yr	+90.8 bp/yr	third digit
bond price response	−9.263%	−9.304%	third digit

Both output moves sit well inside the 13%-wide identification bracket of §8.6, so *the binding limit on this solution is the kink, not the resolution*. That conclusion is the useful output of a convergence study: it says which lever to pull next (calibration, not refinement).

Remark 14.1. This was *not* true before the recalibration described in §8.6. When the constraint was exactly marginal at the rest point, refinement was fighting a Gibbs phenomenon and more nodes bought little. A convergence study's conclusion is conditional on the calibration, and must be re-run after one changes.

14.4 Walras-redundant conditions as independent checks

By Walras's law, one market-clearing condition is implied by the others and the budget constraints. The implementation *drops* the F goods market and the current account from the residual system — including them would make the system singular — and *monitors* them as diagnostics. This is a genuinely independent check: an error in any budget constraint anywhere in the model shows up as a non-zero redundant residual, and no amount of Newton convergence on the retained system can hide it. The practice generalises: *identify the redundancies, drop them from the solve, and watch them.*

In this model the discipline earned its keep. The redundant residual exposed a pre-existing leak from a constant household anchor standing in for a working-capital flow — a bug that the solved system was entirely happy with.

15 Decompositions: making the mechanism auditable

A global solution produces numbers; a paper needs mechanisms. Two decompositions are used, and both are constructed to be *exactly additive* rather than approximately so, which is what makes them auditable.

15.1 The output decomposition and the symmetric two-path split

Production is Cobb–Douglas with predetermined capital, so

$$d\log Y = d\log Z + \alpha\, d\log K + (1-\alpha)\, d\log N. \tag{55}$$

Hours are not free: the GHH labour FOC $\chi N^{1/\text{Frisch}} = w/P$ with the working-capital wedge $w = mc(1-\alpha)(Y/N)/(1+\zeta r^{wc})$ inverts to

$$d\log N = \vartheta\Big[\, d\log Z + \alpha\, d\log K - d\log(1+\zeta r^{wc}) - d\log P\,\Big], \qquad \vartheta = \frac{1}{1/\text{Frisch}+\alpha}. \tag{56}$$

Substituting splits $d\log Y$ into TFP, capital, the wedge, and the price index. The wedge in turn splits by (49) into a *deposit-rate* leg and a *credit-spread* leg — and these move in *opposite* directions, which is exactly the economics the decomposition is there to expose.

Because the two legs enter multiplicatively, their separate contributions are not uniquely defined. The split used is the *symmetric two-path (Shapley) decomposition*: average the contribution of each leg over both orders of introduction. For two factors,

$$\text{contribution}_1 = \tfrac{1}{2}\big[\Delta(1\,|\,\text{base}) + \Delta(1\,|\,2)\big], \tag{57}$$

which is the unique assignment that is symmetric in the factors and sums exactly to the total — the two-player Shapley value. Exact additivity is not an aesthetic preference: it is what makes a decomposition a decomposition rather than an approximation with an unexplained residual.

Every deviation is taken against the *same rules simulated with no shock*, by (54), so the projection's own drift cancels out of the decomposition too.

15.2 The bond-price decomposition

The bank's bond first-order condition is

$$\mathbb{E}[\Omega\,\text{pay}] \;=\; Q\big(\mathbb{E}[\Omega]R + \lambda_b\,\mu\big), \tag{58}$$

which rearranges into three multiplicative legs, hence *exactly additive in logs*:

$$Q \;=\; \underbrace{\frac{\mathbb{E}[\text{pay}]}{R}}_{\text{actuarial}} \times \underbrace{\frac{\mathbb{E}[\Omega\,\text{pay}]}{\mathbb{E}[\Omega]\mathbb{E}[\text{pay}]}}_{\text{risk premium}} \times \underbrace{\frac{\mathbb{E}[\Omega]R}{\mathbb{E}[\Omega]R+\lambda_b\mu}}_{\text{liquidity premium}}. \tag{59}$$

The expected-loss component is isolated by a counterfactual expectation in which a defaulting branch is replaced by the plain no-default one — a clean way to separate "what the payoff is" from "how it is priced". Each leg is recorded per period along the simulated path, so the price response can be attributed quarter by quarter.

Part VII The Other Solver: Sequence-Space Jacobians

16 Linearised solution in sequence space

The second solver answers a different question and does so by entirely different mathematics. It is worth understanding both because the comparison clarifies what each buys.

16.1 The representation

Instead of decision rules on a state space, work directly with *paths*. Stack the unknown sequences $\mathbf{U} = \{U_t\}_{t=0}^{T-1}$ (here 27 variables over $T = 500$ quarters) and the exogenous shock paths $\mathbf{Z}$. Equilibrium requires a vector of target sequences to vanish:

$$\mathbf{H}(\mathbf{U}, \mathbf{Z}) = \mathbf{0}. \tag{60}$$

Linearising around the steady state,

$$H_U \, d\mathbf{U} + H_Z \, d\mathbf{Z} = \mathbf{0} \qquad \Longrightarrow \qquad \boxed{d\mathbf{U} = -H_U^{-1} H_Z \, d\mathbf{Z} \;\equiv\; G \, d\mathbf{Z}.} \tag{61}$$

G is the complete first-order impulse-response operator: once computed, *any* shock path is one matrix–vector product. That is the method's decisive advantage — a policy experiment is a multiplication, not a re-solve.

The blocks H_U and H_Z are assembled from per-block Jacobians by the chain rule along the model's directed graph. For heterogeneous-agent blocks the Jacobian is computed by the "fake-news" algorithm, which exploits the fact that the response to a shock at date s seen from date t depends only on $t - s$ plus a one-period anticipation effect, reducing the cost from $O(T^2)$ expensive backward passes to $O(T)$ cheap ones. The result is exact to first order, not an approximation of the Jacobian.

16.2 A real defect, and why zero-padding is exact

This is a worked example of the kind of problem one meets with library internals, and the reasoning is instructive.

The library's combined-block Jacobian routine seeds its graph traversal *from the shock list* and ends by selecting rows for outputs it actually reached. A target that is a *pure function of the solver's own unknowns* — containing no shock symbol anywhere — is therefore never visited, and is silently dropped from H_Z. Four of the 27 targets are exactly that case (the two price Phillips curves, the terms-of-trade identity and the union-inflation normalisation). The symptom is a dimension mismatch inside the linear solve: a 23-row H_Z against a 27×27 H_U.

The repair restores the missing rows *as zeros*, and the claim is that this is exact rather than an approximation:

Proposition 16.1. *If a target equation h contains no shock symbol Z, then $\partial h / \partial Z = 0$ identically at fixed unknowns, so the corresponding row of H_Z is the zero row.*

The entire content of (61) is a partial derivative at fixed unknowns, so a row that is structurally zero *is* zero. The padding is a restoration, not a regularisation. Two practices make it safe: the padded row names are printed on every solve, so the padding can never go silent; and a repository-wide grep invariant ensures no call site bypasses the wrapper. An alternative fix — folding the four targets into existing equations to get a 25×25 system — was considered and rejected on the ground that it meets the identical defect with smaller numbers, which is worse because it is harder to notice.

> *Transferable lesson.* When a library silently returns an object of the wrong shape, establish whether the missing content is *structurally* determined. If it is, restore it exactly and instrument the restoration. If it is not, the shape error was telling you something about the model.

16.3 When to use which solver

Use the sequence-space solver when	Use the global solver when
the question is a first-order impulse response	the question involves a regime switch
many shock paths must be evaluated (G is reusable)	risk pricing is the mechanism
the constraint's regime does not change	the shock is large relative to the steady state
distributional detail across many agents is needed	the ergodic distribution matters
turnaround time is the binding constraint	the level, not the sign, of a nonlinearity is the object

Part VIII Reference

17 The algorithm, in full

Algorithm 1: global Chebyshev–Smolyak collocation solve

Stage 0 — steady state and anchors

1. Solve the bank franchise fixed point (50) by log-grid sign scan plus Brent, selecting the *least* root (§11.2); recover λ, ω_{ent}; check $\omega_{\text{ent}} > 0$.
2. Stage 1: root-find $\{r_D^k, r_F^k, p\}$ on the two bank net-worth identities plus external balance. Assert the bank identity to 10^{-6} (a large value means fold-blocked targets).
3. Rescale TFP so $Y^{ss} = 1$ exactly (the stage-1 solution is TFP-independent).
4. Stage 2: solve $\{\beta_D, \beta_F\}$ so household saving clears bank deposit supply (EGM household + Young lottery stationary distribution). Impose $\beta_F = \beta_D$ by the symmetric-steady-state doctrine and *check* the F market closes.
5. Calibrate the representative-household transfer anchors so the budget returns $C = C^{ss}$ at the steady state. Iterate to a fixed point rather than solving once, because a trial evaluation can land on a guard.
6. Verify the calibration targets hold exactly; run the Walras goods-market check.

Stage 1 — grid and cold start

7. Build the box: fractional bands on the stocks, $\pm\kappa$ unconditional s.d. on s, absolute bands symmetric about zero on the states that vanish at the steady state. Compose the optional rotation with the shear (§4).
8. Build the Smolyak grid: sparse factor at budget μ over all dimensions but r, tensored with m dense Chebyshev extrema in r. Factorise Φ once.
9. Cold-start every rule at its steady-state value, *except* next-period capital, which tracks the state's own capital so the Jermann inversion is feasible at every node.

Stage 2 — the solve ladder (each rung: short time-iteration warm start, then global Newton)

10. No-default regime at $\pi = 0$.
11. Default regime, warm-started from rung 10, by haircut homotopy $0.85 \to 0.70 \to 0.55 \to$ target.
12. Joint risk-priced solve over both regimes.
13. *(if a policy backstop is on)* cap homotopy $700 \to$ target in 50 bp rungs, in Fischer–Burmeister form, failed rungs retried at half step; convert to Garcia–Zangwill form and re-root.
14. Refinement ladder in r: $3 \to 5 \to 9$ dense nodes, each grid seeded by evaluating the previous solution at the new nodes; the smoothing width walked $10^{-2} \to 10^{-3} \to 10^{-4}$ within each.
15. Stamp on the rule set whether *every* rung rooted, and the worst residual.

Inside one residual evaluation $F(\boldsymbol{\theta})$

16. Unpack $\boldsymbol{\theta}$ to rule values; fit Chebyshev coefficients (exact square solve).
17. For each node and regime: production $\to$ fiscal $\to$ balance sheets $\to$ quadrature over (innovation node $\times$ regime$'$) on the *current* interpolant $\to$ discount kernels $\to$ closed-form multiplier and capital Euler $\to$ bond demands $\to$ policy rule $\to$ wages and dividends $\to$ households $\to$ stack 14 residuals.
18. Append the 6 identity residuals log(guess/implied).
19. Row-scale and ravel. An unevaluable node returns a sentinel so one bad point costs the *step*, not the solve.

Stage 3 — reading

20. Locate the stochastic rest point by zero-shock simulation to 10^{-11}.
21. Compute impulse responses from the rest point, differenced against an unshocked path ((54)); report the fitted and exact reads as a *pair*.
22. Accuracy: simulate $\sim 10^3$ periods, evaluate all 20 conditions at the rule-implied policy, report $\log_{10}$ residual distributions and box-escape frequencies.
23. Decompositions; Walras-redundant diagnostics.

18 Lever reference

Levers are grouped by what they govern. "Failure mode" names what goes wrong when the lever is set badly — which is usually more informative than the nominal description.

Lever	Shipped	Governs / failure mode
Approximation		
`mu`	1	Smolyak total-level budget; cost $O(d^\mu)$. Too low: no cross terms, so interactions are structurally absent — not merely inaccurate
`mu_vec`	`None`	per-dimension caps; still pays the global budget
`refine=(dim,m)`	$(s, 5)$	dense factor in one dimension: degree $m-1$ plus full interaction. The cheap way to buy curvature

Lever	Shipped	Governs / failure mode
S_REFINE	5	shipped resolution (115 points); 9 is the confirmation run
band widths	see §4	coverage vs. wasted degrees. Too narrow: clipping alters effective persistence
S_COVER_SD	2.75	box coverage in unconditional s.d.; invariant to recalibration of σ_s
rot / shear	off / on	box geometry; invariance vs. feasibility (§4)
log/gross fit	on	positivity by construction; removes clips, hence kinks (§12)
Solver		
eps	10^{-6}	finite-difference step. Must sit *below* the smoothing width
cc	1.0	Newton damping
floor_tol	10^{-8}	separates "at the arithmetic floor" from "stalled"
jac_every	1	quadratic rate vs. cost; reuse measured not worth it
TOL_MAXF	10^{-9}	acceptance; four orders below the smallest reported effect
backend	dense	Krylov measured non-convergent (row scaling, §6)
n_jobs	all cores	wall time only; bit-identical to serial
n_gh	5	quadrature exactness $2n-1$; must travel with the solution
warm sweeps	12/20	basin delivery. Refined stages need the longer one
homotopy steps	50 bp	continuation step; halved on failure
TPI_EPS_LADDER	$10^{-2}, 10^{-3}$	smoothing continuation
Nonsmoothness		
tpi_eps (η)	10^{-4}	complementarity smoothing; bias $\eta^2/4g$
_GUARD_EPS	10^{-5}	generic smooth clips
_MU_CAP	0.95	keeps the franchise recursion finite; smoothed
alpha_cap	40	arrests the fold's divergent branch; tunable to be measured
tpi_gz	read/root split	Garcia–Zangwill to store, Fischer–Burmeister to root
Economics that is also numerics		
spread target	100 bp/yr	lifts the ergodic multiplier off the kink; collapses the identification bracket fourfold (§8.6)
leverage target	5.0	sets $\lambda = \alpha/\theta$ directly — the *live* lever
f (exit share)	0.08	sets the fold, hence α; does *not* move the binding margin
ρ_s	0.95	persistence; interacts with duration to produce the repricing
σ_s	0.4455	innovation s.d.; check the quadrature convention (§9)
δ_b	0.056	duration $\approx 1/\delta_b$; inert without persistence
debt_root	0.93	fiscal persistence; a *joint* choice with the debt band
kappa_nfa	0.01	induces stationarity; without it the impulse response is not a well-defined object
ξ (Jermann)	0.50	scales the capital-price leg of equity destruction one-for-one
size_F	8.0	prevents the union deposit rate from cancelling the spread rise; $=$ size$_D$ nests the symmetric model
zeta_wc	1.0	working-capital share; 0 nests the channel off exactly

19 Code map

Approximately twenty percent of this document's purpose is to let a reader find the deployed counterpart of each construct. This table is that index.

File	Theoretical object
Global projection solver	
solver_recursive/state_grid.py	Chebyshev basis (4), nested node families, Smolyak operator (15) in square "new points/new degrees" form, tensor refinement (18), box construction, rotation and shear (§2–4)
solver_recursive/decision_rules.py	the function space: 14 solved + 6 derived rules per regime; the log/gross transforms (§12); the regime table
solver_recursive/point_map.py	one evaluation of $\mathcal{F}$ at a node: closed-form multiplier (36), Fischer–Burmeister (37), smooth guards (39), Garcia–Zangwill (41), product quadrature (45), the 14 residuals
solver_recursive/collocation.py	the global residual $F(\boldsymbol{\theta})$, pack/unpack, identity residuals (28), damped Newton (29), finite-difference Jacobian (30), parallel columns, Newton–Krylov (§6)
solver_recursive/recursive_main.py	time iteration as basin-finder (§5); steady-state anchors; the P-block rotation probe
solver_recursive/recursive_experiment.py	the solve ladder (§7); rest point and double differencing (§13); the fitted/exact bracket
solver_recursive/accuracy.py	ergodic Euler errors and box escapes (§14)
solver_recursive/output_decomposition.py	Shapley output split and the additive bond-price legs
steady_state.py	two-stage steady state; Algorithm 1 Stage 0
blocks/bank.py	the franchise fixed point and its fold (§11.2); target inversion
blocks/household.py, blocks/distribution.py	EGM backward step; Young lottery forward operator
blocks/capital.py, blocks/firms.py, blocks/trade.py, blocks/government.py	Jermann inversion; production; CES trade with size-consistent home bias; the debt root (51)
blocks/fast_kernels.py	JIT kernels with an exact pure-NumPy fallback
Sequence-space solver	
linear_ssj/model/full_model.py	the single block list; $G = -H_U^{-1} H_Z$ (61); the zero-padding result (§6 analogue in Part VII)
linear_ssj/model/equations_*.py	per-country and global blocks
linear_ssj/solve/	steady state and auxiliary calibrations
Shared	
calibration/global_projection.py, calibration/ssj.py	every parameter, with the derived ones derived

File	Theoretical object
run.py	the pipeline and the solver settings of §3

20 Ten transferable lessons

Stated without reference to this model, since they are the part most likely to be useful elsewhere; each is argued where cited.

1. Node placement dominates degree: the Lebesgue constant, not the order, decides whether interpolation converges at all (§2).
2. Sparse grids buy polynomial cost by discarding high-order interactions — so decide *which* interactions the economics needs, because at $\mu = 1$ the missing cross terms are absent structurally, not approximately (§3).
3. Spend resolution where the curvature is: a dense factor in one dimension beat raising the global budget by 36× at fewer points (§3).
4. A residual at zero is not a solution: a change-based stopping rule at tolerance τ admits level error $\tau/(1-\rho)$, and ρ here is 0.99 (§5).
5. Make everything an unknown; the enabling device is the identity residual log(guess/implied), exact at the nodes because square collocation reproduces its nodes (§5).
6. In practice the solver is the homotopy, not the Newton step (§7).
7. Smoothing width and differencing step are one joint choice: $\varepsilon \ll \eta \ll$ economic scale (§8.3).
8. Complementarity reformulations are not interchangeable — min loses slope and stalls, Fischer–Burmeister keeps it, Garcia–Zangwill makes the *stored* rule smooth; one may root in one form and read in another (§8).
9. A risk-pricing model does not rest where its risk-free counterpart does: start at the stochastic rest point *and* difference against an unshocked path, or report a transition between rest points as a response to a shock — here 54% of a reported hump (§13).
10. Report the limit you have: on a constraint's kink the fitted and exactly-cleared reads bracket the answer, and finding the lever that moves the economy off the kink required tracing the calibration map rather than guessing (§8.6).

21 Further reading

Approximation and sparse grids. Trefethen, *Approximation Theory and Approximation Practice* (2013) — the definitive modern treatment of Chebyshev interpolation, including Theorem 2.1 and the Lebesgue-constant results. Smolyak (1963) for the original construction; Barthelmann, Novak and Ritter (2000) for error bounds; Krueger and Kubler (2004) and Judd, Maliar, Maliar and Valero (2014) for the economic implementations, the latter including anisotropic and adaptive variants. Bungartz and Griebel (2004) is the standard survey.

Projection and global solution in economics. Judd, *Numerical Methods in Economics* (1998), chapters 6, 11 and 17. Fernández-Villaverde, Rubio-Ramírez and Schorfheide (2016) for the handbook survey of solution and estimation methods. Maliar and Maliar (2014) on the curse of dimensionality.

Newton methods and continuation. Kelley, *Iterative Methods for Linear and Nonlinear Equations* (1995) — line searches, Shamanskii, and the Newton–Krylov conditioning issues of §6. Allgower and Georg, *Numerical Continuation Methods* (1990) for the homotopy theory behind §7.

Complementarity. Facchinei and Pang, *Finite-Dimensional Variational Inequalities and Complementarity Problems* (2003) for Fischer–Burmeister and its smoothing; Garcia and Zangwill, *Pathways to Solutions, Fixed Points and Equilibria* (1981) for the single-variable reformulation of §8.4.

Heterogeneous agents and sequence space. Carroll (2006) on the endogenous grid method; Young (2010) for the lottery distribution operator; Auclert, Bardóczy, Rognlie and Straub (2021) for sequence-space Jacobians and the fake-news algorithm of Part VII.

The economics being solved. Gertler and Karadi (2011) and Gertler and Kiyotaki (2010) for the intermediary block and the leverage constraint; Bocola (2016), "The Pass-Through of Sovereign Risk", for the mechanism, the solution design this solver follows, and the closed-form multiplier of (36); Hatchondo and Martinez (2009) for the perpetuity bond; Neumeyer and Perri (2005) for the working-capital wedge (49); Schmitt-Grohé and Uribe (2003) for the stationarity-inducing premium; Jermann (1998) for the capital adjustment cost.

Every numerical magnitude quoted in this document is a measurement recorded in the repository's own development history, re-verified against the live code and calibration at the time of writing. The grid point counts, interpolation-exactness checks and all four figures were computed directly from the deployed grid constructor. Where a number brackets rather than identifies a quantity — notably the impact output response of §8.6 — it is reported as a bracket.